

Package ‘hydrographr’

April 27, 2026

Type Package

Title Scalable Hydrographic Data Processing in R

Date 2026

Version 1.6.0

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Description Scalable hydrographic geospatial data processing tools using open-source command-line utilities and a remote spatial database accessible via an OGC API. The package provides functions to download the Hydrography90m data (<https://essd.copernicus.org/articles/14/4525/2022/>), process and extract hydrographic information, and assess network distances and connectivity. A dedicated set of API functions serves as a user-friendly R wrapper around an OGC API-compliant server hosting the Hydrography90m dataset in a PostgreSQL/PostGIS database, enabling computing-intensive spatial operations to be performed server-side without local data downloads. While the local processing functions are tailored towards the Hydrography90m data, they can also be applied to other data for tasks such as cropping and merging raster and vector data, point-raster extraction, raster reclassification, and data aggregation. Local processing depends on GDAL/OGR, GRASS-GIS and AWK in a Linux environment. Windows users need to first activate the Windows Subsystem for Linux (WSL) feature. Please see the installation guide at <https://glowabio.github.io/hydrographr/articles/hydrographr.html> and instructions for use with other stream networks at https://glowabio.github.io/hydrographr/articles/example_other_stream_networks.html.

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URL <https://glowabio.github.io/hydrographr/>

BugReports <https://github.com/glowabio/hydrographr/issues>

Encoding UTF-8

LazyData true

RoxygenNote 7.3.3

Imports processx (>= 3.7.0),
 data.table (>= 1.14.2),
 tidyr (>= 1.2.1),
 dplyr (>= 1.0.10),
 stringi (>= 1.7.8),
 stringr (>= 1.4.1),
 rlang (>= 1.0.6),
 DBI (>= 1.1.3),
 RSQLite (>= 2.2.19),
 terra (>= 1.6-41),
 sf (>= 1.0-9),
 parallel (>= 4.2.2),
 doParallel (>= 1.0.17),
 foreach (>= 1.5.2),
 future (>= 1.29.0),
 doFuture (>= 0.12.2),
 future.apply (>= 1.10.0),
 memuse (>= 4.2-2),
 igraph (>= 1.3.5),
 magrittr (>= 2.0.3),
 methods (>= 4.3.0)

Suggests testthat

Collate 'api_get_snapped_points.R'
 'check_tiles_filesize.R'
 'crop_to_extent.R'
 'download_env90m_tables.R'
 'download_test_data.R'
 'download_tiles.R'
 'download_tiles_base.R'
 'extract_from_gpkg.R'
 'extract_ids.R'
 'extract_lake_ids.R'
 'extract_partial_stream_network.R'
 'extract_zonal_stat.R'
 'get_all_upstream_distances.R'
 'get_buffer_along_the_network.R'
 'get_catchment_graph.R'
 'get_centrality.R'
 'get_distance.R'
 'get_distance_parallel.R'
 'get_distance_graph.R'
 'get_lake_intersection.R'
 'get_lake_catchment.R'
 'get_modelfit_table.R'
 'get_pfafstetter_basins.R'
 'get_predict_table.R'
 'get_regional_unit_id.R'
 'get_segment_neighbours.R'

'get_tile_id.R'
 'get_upstream_catchment.R'
 'get_upstream_variable.R'
 'utils.R'
 'merge_tile.R'
 'read_geopackage.R'
 'reclass_raster.R'
 'report_no_data.R'
 'set_no_data.R'
 'snap_to_network.R'
 'snap_to_subc_segment.R'
 'split_table.R'
 'api_poll_job.R'
 'api_get_basin_polygon.R'
 'api_get_snapped_points_cascade.R'
 'api_get_upstream_catchment.R'
 'crop_vector_to_extent.R'
 'api_get_stream_segments.R'
 'api_get_ids.R'

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api_get_basin_polygon *Get Basin Polygon(s)*

Description

This function returns whole basin polygon(s) as an 'sf' object. Two input modes: 1. One or more basin_ids (vector of integers) 2. A data.frame with point coordinates (retrieves basin polygons via lon/lat)

Usage

```
api_get_basin_polygon(
  basin_id = NULL,
  points_df = NULL,
  colname_lat = "latitude",
  colname_lon = "longitude",
  colname_site_id = "site_id",
  geometry_only = FALSE,
  comment = NULL
)
```

Arguments

basin_id	Integer or integer vector. The ID(s) of the basin(s) to retrieve. If NULL, must provide 'points_df'.
points_df	A data.frame with columns for longitude, latitude, and site_id. Used if 'basin_id' is NULL. Each row is sent as a lon/lat point to the API.
colname_lat	Character. Name of latitude column in 'points_df'. Default: "latitude".

colname_lon	Character. Name of longitude column in 'points_df'. Default: "longitude".
colname_site_id	Character. Name of site ID column in 'points_df'. Default: "site_id".
geometry_only	Logical. If 'TRUE', returns only geometry without attributes. Defaults to 'FALSE'.
comment	Character. Optional comment for API logging.

Details

Retrieves whole basin polygon boundary/boundaries from the GeoFRESH API. Can accept either basin_id(s) directly or a data.frame of point coordinates (lon, lat, site_id). All requests are sent as JSON to the API.

Value

An 'sf' object representing the basin polygon(s). If multiple basins are retrieved, returns all unique basins.

See Also

Other ocgapi: [api_get_ids\(\)](#), [api_get_snapped_points\(\)](#), [api_get_snapped_points_cascade\(\)](#), [api_get_stream_segments\(\)](#), [api_get_upstream_catchment\(\)](#)

Examples

```
## Not run:
# Method 1: Using a single basin_id
basin_sf <- api_get_basin_polygon(
  basin_id = 1288419,
  geometry_only = FALSE
)

# Method 1b: Using a vector of basin_ids
basins_sf <- api_get_basin_polygon(
  basin_id = c(1288419, 1288420, 1288421)
)

# Method 2: Using a data.frame of points
pts <- data.frame(
  site_id = c("site_a", "site_b"),
  longitude = c(8.6, 9.1),
  latitude = c(53.5, 52.3)
)
basins_sf <- api_get_basin_polygon(points_df = pts)

# Visualize
library(leaflet)
leaflet(basins_sf) %>%
  addProviderTiles("CartoDB.Positron") %>%
  addPolygons(color = "blue", fillOpacity = 0.3)
```

```
## End(Not run)
```

api_get_ids

Get Subcatchment/Geographic IDs

Description

Retrieves subcatchment IDs and related geographic identifiers from the GeoFRESH API. Three modes cover distinct spatial queries:

Usage

```
api_get_ids(
  points = NULL,
  basin_ids = NULL,
  subc_ids = NULL,
  mode = "basin",
  min_strahler = NULL,
  colname_lon = "longitude",
  colname_lat = "latitude",
  colname_site_id = "site_id",
  colname_subc_id = NULL,
  comment = NULL,
  force_async = TRUE,
  poll_interval = 10,
  max_wait = 3600
)
```

Arguments

points	Data.frame. Input sites. Used in all three modes: - mode="basin": must contain longitude and latitude columns (and optionally site_id). Can alternatively supply 'basin_ids' or 'subc_ids' directly. - mode="upstream": must contain longitude, latitude, and site_id columns. - mode="local": must contain either longitude + latitude columns, or a subc_id column. Returns one row of IDs per input row.
basin_ids	Integer or vector. Basin IDs (mode="basin" only). Alternative to 'points' or 'subc_ids'.
subc_ids	Integer or vector. Subcatchment IDs (mode="basin" only). Alternative to 'points' or 'basin_ids'.
mode	Character. One of "basin", "upstream", or "local". Default: "basin".
min_strahler	Integer. Minimum Strahler order filter (mode="basin" and mode="upstream" only). Default: NULL.
colname_lon	Character. Name of the longitude column in 'points'. Default: "longitude".
colname_lat	Character. Name of the latitude column in 'points'. Default: "latitude".

colname_site_id	Character. Name of the site ID column in 'points'. Default: "site_id".
colname_subc_id	Character. Name of an existing subcatchment ID column in 'points' (mode="local" only). If provided, the API uses these directly instead of looking up from coordinates. Default: NULL.
comment	Character. Optional comment for API logging. Default: NULL.
force_async	Logical. Force async (TRUE) or sync (FALSE) mode. Default: TRUE.
poll_interval	Numeric. Seconds between status checks. Default: 10.
max_wait	Numeric. Max seconds to wait for async job. Default: 3600.

Details

- **"basin"**: returns all subcatchment IDs contained within the basin(s) of the given locations. - **"upstream"**: returns all subcatchment IDs upstream of each input point. - **"local"**: returns the local identifiers (subc_id, basin_id, reg_id) for each input point or subcatchment — i.e. the reverse lookup.

Value

- mode="basin": Integer vector of all subcatchment IDs in the basin(s). - mode="upstream": Data.frame with columns 'site_id', 'subc_id', 'upstream_id' (one row per upstream subcatchment per site). - mode="local": Data.frame with one row per input point, containing 'subc_id', 'basin_id', and 'reg_id'.

See Also

Other ocgapi: [api_get_basin_polygon\(\)](#), [api_get_snapped_points\(\)](#), [api_get_snapped_points_cascade\(\)](#), [api_get_stream_segments\(\)](#), [api_get_upstream_catchment\(\)](#)

Examples

```
## Not run:
# Basin mode: get all subcatchments in a basin
result <- api_get_ids(basin_ids = 1288419, mode = "basin")

# Basin mode: from snapped points dataframe
result <- api_get_ids(
  points      = all_snapped,
  colname_lon = "longitude_snapped",
  colname_lat = "latitude_snapped",
  colname_site_id = "site_id",
  mode        = "basin"
)

# Upstream mode: all upstream subcatchments per site
sites <- data.frame(site_id = "S1", longitude = 9.93, latitude = 54.70)
result <- api_get_ids(points = sites, mode = "upstream")

# Local mode: get subc_id, basin_id, reg_id for each point
```

```

sites <- data.frame(longitude = c(10.7, 9.9), latitude = c(53.5, 54.7))
result <- api_get_ids(points = sites, mode = "local")

# Local mode: from existing subc_ids column
result <- api_get_ids(
  points      = my_df,
  colname_subc_id = "subc_id",
  mode        = "local"
)

## End(Not run)

```

api_get_snapped_points

Snap Points to the River Network

Description

Snaps input point locations to the nearest stream segment at or above a given Strahler order, using the GeoFRESH API.

Usage

```

api_get_snapped_points(
  data,
  colname_lon = "longitude",
  colname_lat = "latitude",
  colname_site_id = "site_id",
  min_strahler = NULL,
  add_distance = TRUE,
  force_async = NULL,
  async_threshold = 500,
  poll_interval = 10,
  max_wait = 3600
)

```

Arguments

data	Data.frame. Must contain columns for longitude, latitude, and a site identifier.
colname_lon	Character. Name of the longitude column. Default: "longitude".
colname_lat	Character. Name of the latitude column. Default: "latitude".
colname_site_id	Character. Name of the site ID column. Default: "site_id".
min_strahler	Integer. Minimum Strahler order to snap to. Required.
add_distance	Logical. If TRUE, includes a 'distance_metres' column with the snapping distance. Default: TRUE.

force_async	Logical. If NULL (default), async mode is used automatically for datasets $\geq$ 'async_threshold'. Set TRUE or FALSE to force a mode regardless of dataset size.
async_threshold	Integer. Number of points above which async mode is used automatically. Default: 500.
poll_interval	Numeric. Seconds between status checks when polling async jobs. Default: 10.
max_wait	Numeric. Maximum seconds to wait for async job completion. Default: 3600.

Value

Data.frame with one row per snapped point, containing: - All original columns from 'data' - 'longitude_snapped', 'latitude_snapped': snapped coordinates - 'subc_id': subcatchment ID of the snapped segment - 'strahler': Strahler order of the snapped segment - 'distance_metres': distance from original to snapped point (if 'add_distance = TRUE')

See Also

Other ocgapi: [api_get_basin_polygon\(\)](#), [api_get_ids\(\)](#), [api_get_snapped_points_cascade\(\)](#), [api_get_stream_segments\(\)](#), [api_get_upstream_catchment\(\)](#)

Examples

```
## Not run:
sites <- data.frame(
  site_id = c("S1", "S2", "S3"),
  longitude = c(9.931555, 9.921555, 9.941555),
  latitude = c(54.695070, 54.295070, 54.495070)
)

# Snap to streams with Strahler  $\geq$  4
result <- api_get_snapped_points(data = sites, min_strahler = 4)

# Non-default column names
result <- api_get_snapped_points(
  data = sites,
  colname_lon = "decimalLongitude",
  colname_lat = "decimalLatitude",
  colname_site_id = "gbifID",
  min_strahler = 4
)

## End(Not run)
```

api_get_snapped_points_cascade

Snap Points to Stream Network (Cascade Workflow) - JSON Input

Description

Snaps points to stream network across descending Strahler orders using data.frame input. Points exceeding distance threshold at higher orders are re-snapped to lower orders until successful or all levels exhausted.

Usage

```
api_get_snapped_points_cascade(
  data,
  colname_lon = "longitude",
  colname_lat = "latitude",
  colname_site_id = "site_id",
  strahler_seq = c(4, 3, 2),
  distance_threshold = 150,
  async_threshold = 500,
  poll_interval = 10,
  max_wait = 7200
)
```

Arguments

data	A data.frame containing point data with coordinates and site IDs.
colname_lon	Character. Longitude column name in data.frame. Default: "longitude".
colname_lat	Character. Latitude column name in data.frame. Default: "latitude".
colname_site_id	Character. Site ID column name in data.frame. Default: "site_id".
strahler_seq	Integer vector. Descending Strahler orders to try (e.g., 'c(4, 3, 2)'). Higher values tried first.
distance_threshold	Numeric. Maximum snapping distance in meters. Points exceeding this are re-snapped to next lower Strahler level. Default: '150'.
async_threshold	Numeric. Number of points threshold for auto-async. Default: '500'.
poll_interval	Numeric. Seconds between async job status checks. Default: '10'.
max_wait	Numeric. Maximum seconds to wait for job completion. Default: '7200' (2 hours).

Details

****Workflow per Strahler level:****

1. Convert data.frame to GeoJSON and snap to current Strahler level
2. Poll until job completes (if async)
3. Separate points within vs exceeding distance threshold
4. Store successful snaps, prepare failed snaps for next level
5. Re-snap failures from original coordinates at next lower Strahler

Points always snap from original coordinates to ensure each attempt starts fresh.

Value

A data.frame with successfully snapped points containing:

site_id Site identifier from input data.

subc_id Subcatchment ID of snapped location.

strahler Strahler order of stream segment.

longitude_snapped, latitude_snapped Snapped coordinates.

longitude_original, latitude_original Original coordinates.

distance_metres Snapping distance in meters.

... Any other columns from original data.

Returns empty data.frame if no points successfully snapped.

See Also

Other ocgapi: [api_get_basin_polygon\(\)](#), [api_get_ids\(\)](#), [api_get_snapped_points\(\)](#), [api_get_stream_segments\(\)](#), [api_get_upstream_catchment\(\)](#)

Examples

```
## Not run:
# Create sample data
sites <- data.frame(
  site_id = c("site_1", "site_2", "site_3"),
  longitude = c(9.931555, 9.921555, 9.941555),
  latitude = c(54.695070, 54.295070, 54.495070),
  species = c("Trout", "Salmon", "Pike")
)

# Run cascade
result <- api_get_snapped_points_cascade(
  data = sites,
  strahler_seq = c(5, 4, 3),
  distance_threshold = 200
)
```

```
# Large dataset (auto-async)
gbif_data <- data.table::fread("my_fish_data.csv")
result <- api_get_snapped_points_cascade(
  data = gbif_data,
  colname_lon = "decimalLongitude",
  colname_lat = "decimalLatitude",
  colname_site_id = "gbifID",
  strahler_seq = c(4, 3, 2)
)

## End(Not run)
```

api_get_stream_segments

Get Stream Segments of a Basin or Upstream of a Point

Description

This function returns stream segments as an 'sf' object with optional filtering by Strahler order. Two modes of operation:

****Basin mode**** (default, 'upstream = FALSE'): Returns all stream segments *within* a specified basin. The basin can be identified by one of: 'basin_id', 'subc_id', or coordinates ('lon' + 'lat').

****Upstream mode**** ('upstream = TRUE'): Returns all stream segments *upstream of* a specified location. Only accepts 'subc_id' or coordinates ('lon' + 'lat'); 'basin_id' is not compatible with upstream mode. All requests are sent as JSON to the API.

Usage

```
api_get_stream_segments(
  basin_id = NULL,
  subc_id = NULL,
  lon = NULL,
  lat = NULL,
  upstream = FALSE,
  geometry_only = FALSE,
  min_strahler = NULL,
  add_segment_ids = NULL,
  add_upstream_ids = FALSE
)
```

Arguments

basin_id	Integer. The ID of the basin. Only valid in basin mode ('upstream = FALSE'). One of 'basin_id', 'subc_id', or ('lon' + 'lat') must be provided. Optional. Default: NULL.
----------	--

subc_id	Integer. A sub-catchment ID from which the basin is inferred (basin mode) or from which to trace upstream (upstream mode). Optional. Default: NULL.
lon	Numeric. Longitude of a point (used together with 'lat'). Works in both basin mode and upstream mode. Optional. Default: NULL.
lat	Numeric. Latitude of a point (used together with 'lon'). Latitude must be between -90 and 90. Optional. Default: NULL.
upstream	Logical. If 'FALSE' (default), returns segments within the basin. If 'TRUE', returns segments upstream of the specified location. Requires 'subc_id' or ('lon' + 'lat'); 'basin_id' is not supported in upstream mode. Default: FALSE.
geometry_only	Logical. If 'TRUE', returns only the geometry of the stream segments without associated attributes. Defaults to 'FALSE'.
min_strahler	Integer (optional). The minimum Strahler stream order to include. Streams below this order will be excluded. Defaults to 'NULL' (no filtering).
add_segment_ids	Logical (optional). If 'TRUE', segment IDs are added to the output. Only used in basin mode. Defaults to 'NULL' (omitted from request).
add_upstream_ids	Logical (optional). If 'TRUE', upstream sub-catchment IDs are added to the output. Only used in upstream mode. Defaults to 'FALSE'.

Details

Sends a request to the GeoFRESH API to retrieve stream segments belonging to a given basin, or all upstream segments of a specific location.

Value

An 'sf' object representing the stream segments. Geometry type: LINESTRING.

See Also

Other ocgapi: [api_get_basin_polygon\(\)](#), [api_get_ids\(\)](#), [api_get_snapped_points\(\)](#), [api_get_snapped_points_ca](#), [api_get_upstream_catchment\(\)](#)

Examples

```
## Not run:
# === BASIN MODE (upstream = FALSE) ===

# Using basin_id
segm_sf <- api_get_stream_segments(
  basin_id = 1288419,
  geometry_only = FALSE,
  min_strahler = 4,
  add_segment_ids = TRUE
)

# Using subc_id
```

```

segm_sf <- api_get_stream_segments(
  subc_id = 506586041,
  geometry_only = TRUE
)

# Using lon/lat
segm_sf <- api_get_stream_segments(
  lon = 8.278198242187502,
  lat = 53.54910661890981,
  geometry_only = TRUE
)

# === UPSTREAM MODE (upstream = TRUE) ===

# Using subc_id to get upstream segments
upstream_sf <- api_get_stream_segments(
  subc_id = 506586041,
  upstream = TRUE,
  geometry_only = FALSE,
  min_strahler = 1
)

# Using lon/lat to get upstream segments (Sarantaporos outlet)
upstream_sf <- api_get_stream_segments(
  lon = 20.538704,
  lat = 40.113735,
  upstream = TRUE,
  geometry_only = FALSE
)

# Visualize with leaflet
library(leaflet)
leaflet(upstream_sf) %>%
  addProviderTiles("CartoDB.Positron") %>%
  addPolylines(color = "blue")

## End(Not run)

```

api_get_upstream_catchment

Get Upstream Catchment Polygon (Dissolved)

Description

This function returns a single dissolved polygon representing the merged area of all sub-catchments upstream of a specified location. The location can be identified by one of: 'subc_id', a pair of coordinates ('lon' + 'lat'), or a GeoJSON 'point'. All inputs are sent as JSON.

Usage

```
api_get_upstream_catchment(
  subc_id = NULL,
  lon = NULL,
  lat = NULL,
  point = NULL,
  add_upstream_ids = FALSE,
  comment = NULL
)
```

Arguments

subc_id	Integer. A sub-catchment ID from which to trace upstream. One of 'subc_id', ('lon' + 'lat'), or 'point' must be provided.
lon	Numeric. Longitude of a point (used together with 'lat').
lat	Numeric. Latitude of a point (used together with 'lon').
point	List. A GeoJSON Point geometry or Feature. Alternative to providing 'lon' + 'lat'. Optional.
add_upstream_ids	Logical. If 'TRUE', upstream sub-catchment IDs are included in the output. Defaults to 'FALSE'.
comment	Character. Optional comment for API logging.

Details

Sends a request to the GeoFRESH API to retrieve the dissolved upstream catchment polygon for a given point or sub-catchment and returns it as an 'sf' object.

Value

An 'sf' object representing the dissolved upstream catchment polygon.

See Also

Other `ocgapi`: [api_get_basin_polygon\(\)](#), [api_get_ids\(\)](#), [api_get_snapped_points\(\)](#), [api_get_snapped_points_ca](#), [api_get_stream_segments\(\)](#)

Examples

```
## Not run:
# Using lon/lat (Sarantaporos outlet)
catchment_sf <- api_get_upstream_catchment(
  lon = 20.538704,
  lat = 40.113735
  add_upstream_ids = TRUE
)

# Using subc_id
```

```

catchment_sf <- api_get_upstream_catchment(
  subc_id = 506586041
)

# Using a GeoJSON point
catchment_sf <- api_get_upstream_catchment(
  point = list(
    type = "Feature",
    geometry = list(
      type = "Point",
      coordinates = c(20.538704, 40.113735)
    )
  )
)

library(leaflet)
leaflet(catchment_sf) %>%
  addProviderTiles("CartoDB.Positron") %>%
  addPolygons(color = "blue")

## End(Not run)

```

api_poll_job

Poll status of an async job

Description

Poll status of an async job

Usage

```

api_poll_job(
  jobID,
  wait = TRUE,
  poll_interval = 5,
  max_wait = 72000,
  base_url = "https://aqua.igb-berlin.de/pygeoapi"
)

```

Arguments

jobID	Character. The job ID returned from an async API call.
wait	Logical. If TRUE, poll until complete. If FALSE, check once.
poll_interval	Numeric. Seconds between checks (default: 5).
max_wait	Numeric. Maximum wait time in seconds (default: 600).
base_url	Character. Base URL for the pygeoapi instance.

Value

List with job status and href (if complete)

crop_to_extent	<i>Crop raster to extent</i>
----------------	------------------------------

Description

This function crops an input raster layer (.tif) given a bounding box (xmin, ymin, xmax, ymax coordinates, or a spatial object from which to extract a bounding box) or to the boundary of a polygon vector layer (cutline source). The cropping is performed directly on disk, i.e. the input layer does not need to be loaded into R. The output is always written to disk, and can be optionally loaded into R as a SpatRaster (terra package) object (using read = TRUE).

Usage

```
crop_to_extent(
  raster_layer,
  vector_layer = NULL,
  bounding_box = NULL,
  out_dir,
  file_name,
  compression = "low",
  bigtiff = TRUE,
  read = TRUE,
  quiet = TRUE
)
```

Arguments

raster_layer	character. Full path to the input raster .tif layer.
vector_layer	character. Full path to a vector layer that is used as a cutline data source (similar to a mask operation).
bounding_box	numeric vector of the coordinates of the corners of a bounding box (xmin, ymin, xmax, ymax), SpatRaster, SpatVector, or other spatial object.
out_dir	character. The directory where the output will be stored.
file_name	character. Name of the cropped output raster .tif file.
compression	character. Compression of the written output file. Compression levels can be defined as "none", "low", or "high". Default is "low".
bigtiff	logical. Define whether the output file is expected to be a BIGTIFF (file size larger than 4 GB). If FALSE and size > 4GB no file will be written. Default is TRUE.
read	logical. If TRUE, the cropped raster .tif layer gets read into R. If FALSE, the layer is only stored on disk. Default is TRUE.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Value

The function returns always a .tif raster file written to disk. Optionally, a SpatRaster (terra object) can be loaded into R with read = TRUE.

Author(s)

Yusdiel Torres-Cambas

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Define full path to the input raster .tif layer and vector layer
spi_raster <- paste0(my_directory, "/hydrography90m_test_data",
                     "/spi_1264942.tif")
basin_vector <- paste0(my_directory, "/hydrography90m_test_data",
                      "/basin_59.gpkg")

# Crop the Stream Power Index to the basin
spi_basin <- crop_to_extent(raster_layer = spi_raster,
                           vector_layer = basin_vector,
                           out_dir = my_directory,
                           file_name = "spi_basin_cropped.tif",
                           read = TRUE)
```

crop_vector_to_extent *Crop vector to extent*

Description

This function crops an input vector layer (.shp, .gpkg, .geojson) given a bounding box (xmin, ymin, xmax, ymax coordinates, or a spatial object from which to extract a bounding box) or to the boundary of a polygon vector layer (cutline source). The cropping is performed directly on disk using ogr2ogr, i.e. the input layer does not need to be loaded into R. The output is always written to disk, and can be optionally loaded into R as an sf object (using read = TRUE).

Usage

```
crop_vector_to_extent(
  vector_layer,
  clip_layer = NULL,
  bounding_box = NULL,
  out_dir,
  file_name,
  read = TRUE,
  quiet = TRUE
)
```

Arguments

vector_layer	character. Full path to the input vector layer (.shp, .gpkg, .geojson).
clip_layer	character. Full path to a polygon vector layer used as the clipping boundary (similar to a mask operation). Optional.
bounding_box	numeric vector of the coordinates of the corners of a bounding box (xmin, ymin, xmax, ymax), SpatRaster, SpatVector, or other spatial object.
out_dir	character. The directory where the output will be stored.
file_name	character. Name of the cropped output vector file. Supported formats: .gpkg, .shp, .geojson.
read	logical. If TRUE, the cropped vector layer gets read into R as an sf object. If FALSE, the layer is only stored on disk. Default is TRUE.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Value

The function always writes a vector file to disk. Optionally, an sf object can be loaded into R with `read = TRUE`.

Note

Shapefile output (.shp) truncates column names to 10 characters due to format limitations. Use .gpkg or .geojson to preserve full column names.

Author(s)

Afroditi Grigoropoulou

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Define full path to the input vector layer
basin_vector <- paste0(my_directory, "/hydrography90m_test_data",
                       "/basin_59.gpkg")

# Crop a vector layer to a bounding box
cropped_sf <- crop_vector_to_extent(
  vector_layer = basin_vector,
  bounding_box = c(8.5, 42.24, 8.8, 42.3),
  out_dir = my_directory,
  file_name = "basin_cropped.gpkg",
  read = TRUE
)
```

download_env90m_tables

Download Environment90m tables

Description

The various ‘download_something_tables()’ functions allow to retrieve the Environment90m variable names and download data of the Environment90m datasets, which are split into 20°x20° tiles.

There are basically 3 usages:

- (1) If the functions are called without arguments (i.e. without specifying variable names and tiles), the available variable names are returned.
- (2) If a subset of variables and tile IDs are specified, the download size of the resulting download will be computed.
- (3) If a subset of variables and tile IDs are specified, and ‘download’ is set to ‘TRUE’, the requested tables will be downloaded and either left as zipped files or unzipped to text files.

Multiple regular tiles, e.g. belonging to regional units, can be downloaded in a single request. The tile IDs can be obtained using the function [get_tile_id()].

Usage

```
download_soil_tables(
  subset = NULL,
  tile_ids = NULL,
  download = FALSE,
  download_dir = ".",
  file_format = "txt",
  delete_zips = TRUE,
  ignore_missing = FALSE,
  tempdir = NULL,
  quiet = FALSE
)
```

```
download_flo1k_tables(
  subset = NULL,
  tile_ids = NULL,
  download = FALSE,
  download_dir = ".",
  file_format = "txt",
  delete_zips = TRUE,
  ignore_missing = FALSE,
  tempdir = NULL,
  quiet = FALSE
)
```

```
download_cg iar_tables(
```

```
subset = NULL,  
tile_ids = NULL,  
download = FALSE,  
download_dir = ".",  
file_format = "txt",  
delete_zips = TRUE,  
ignore_missing = FALSE,  
tempdir = NULL,  
quiet = FALSE  
)  
  
download_merit_dem_tables(  
  subset = NULL,  
  tile_ids = NULL,  
  download = FALSE,  
  download_dir = ".",  
  file_format = "txt",  
  delete_zips = TRUE,  
  ignore_missing = FALSE,  
  tempdir = NULL,  
  quiet = FALSE  
)  
  
download_hydrography90m_tables(  
  subset = NULL,  
  tile_ids = NULL,  
  download = FALSE,  
  download_dir = ".",  
  file_format = "txt",  
  delete_zips = TRUE,  
  ignore_missing = FALSE,  
  tempdir = NULL,  
  quiet = FALSE  
)  
  
download_observed_climate_tables(  
  subset = NULL,  
  tile_ids = NULL,  
  download = FALSE,  
  download_dir = ".",  
  file_format = "txt",  
  delete_zips = TRUE,  
  ignore_missing = FALSE,  
  tempdir = NULL,  
  quiet = FALSE  
)  
  
download_projected_climate_tables(  
  subset = NULL,  
  tile_ids = NULL,  
  download = FALSE,  
  download_dir = ".",  
  file_format = "txt",  
  delete_zips = TRUE,  
  ignore_missing = FALSE,  
  tempdir = NULL,  
  quiet = FALSE  
)
```

```

base_vars = NULL,
time_periods = NULL,
models = NULL,
scenarios = NULL,
versions = NULL,
subset = NULL,
tile_ids = NULL,
download = FALSE,
download_dir = ".",
file_format = "txt",
delete_zips = TRUE,
ignore_missing = FALSE,
tempdir = NULL,
quiet = FALSE
)

download_landcover_tables(
  base_vars = NULL,
  years = NULL,
  subset = NULL,
  tile_ids = NULL,
  download = FALSE,
  download_dir = ".",
  file_format = "txt",
  delete_zips = TRUE,
  ignore_missing = FALSE,
  tempdir = NULL,
  quiet = FALSE
)

```

Arguments

subset	Vector of the variable names that should be downloaded (or string "ALL" for all available variables).
tile_ids	Vector containing all tile ids of the tiles (e.g. "h10v04") that should be downloaded, or whose download availability and size should be checked (or string "ALL" for all available tiles).
download	logical. If TRUE, and if 'tile_ids' is specified, the files will be downloaded from the IGB server. If FALSE, and if 'tile_ids' is specified, the download size will be computed. If FALSE, and if 'tile_ids' is NULL, the variable names will be returned to the user.
download_dir	Directory where the downloads should be stored. Defaults to the current working directory ".". Ignored if 'download=FALSE'.
file_format	File format of the tables, either "txt" or "zip". If "txt", then the zipped tables are unzipped. If "zip", the downloaded zipped files are left as they are. Default is "txt", which means that the zip files are unzipped after downloading. Note that this will take more space on disk than zips.

delete_zips	logical. boolean If 'FALSE', the downloaded zip files are not deleted after unzipping. Defaults to TRUE. This is ignored if you request file format zip.
ignore_missing	logical. What to do if some of the requested variables and/or tile_ids are not available, which is most frequently caused by a typo the variable name. If TRUE, the missing or misspelled ones are ignored while the others are downloaded (and a warning is given out). If FALSE, the function will fail to allow the user to check the variable names and their spelling. Defaults to FALSE.
tempdir	Optional (rarely needed). Path to the directory where to store/look for the temporary various file size tables for the various Environment90m datasets, which are required and downloaded by the functions. If not passed, defaults to the output of [base::tempdir()].
quiet	logical. If FALSE, informative messages will be printed. Default is FALSE.
base_vars	(Only in 'download_projected_climate_tables()' and 'download_landcover_tables()') Vector of the desired base variables, e.g. the landcover variable "c20_1992" can be expressed as base variable "c20" and year "1992".
time_periods	(Only in 'download_projected_climate_...') Vector of the desired time periods (leave 'NULL' or specify "ALL" for all available time periods).
models	(Only in 'download_projected_climate_...') Vector of the desired models (leave 'NULL' or specify "ALL" for all available models).
scenarios	(Only in 'download_projected_climate_...') Vector of the desired scenarios (leave 'NULL' or specify "ALL" for all available scenarios).
versions	(Only in 'download_projected_climate_...') Vector of the desired versions (leave 'NULL' or specify "ALL" for all available versions). As of January 2025, the only available version is "V.2.1".
years	(Only in 'download_landcover_...') Vector of the desired years (leave 'NULL' or specify "ALL" for all available years).

Details

In the following table you can find all the variables included in the Environment90m dataset. The column "Variable" includes the variable names that should be used as an input in the parameter "variable" of the function. Likewise, the column "File format" contains the input that should be given to the "file_format" parameter.

The Environment90m dataset comprises data from the landcover dataset ESA Land Cover (esa_cci_landcover_v2_1_1), CHELSA v2.1 (chelsa_bioclim_v2_1), SOILGRIDS (soilgrids250m_v2_0) and Hydrography90m (hydrography90m_v1_0).

For visualisations the available tiles, and for details on the variables of the Hydrography90m dataset, please refer to https://hydrography.org/hydrography90m/hydrography90m_layers/.

For details on the bioclimatic variables, especially for details of the scale and unit of the values, please refer to <http://chelsa-climate.org/>.

For details on the ESA Land Cover variables, please refer to <https://www.climatologylab.org/terraclimate.html>. Please note that some values in this dataset are aggregated from similar classes (see Environment90m publication).

For details on the Soil data, please refer to <https://soilgrids.org>.

Value

A named list of: * variable names, * components of variable names (if applies), * the dataset name, * the requested tile_ids (if applies), * the download size (if 'tile_id' is specified), * the path where downloaded files are stored to (if 'download=TRUE'), * etc.

Functions

- `download_soil_tables()`: Download SOILGRIDS tables (`soilgrids250m_v2_0`)
- `download_flow_tables()`: Download flow tables (`flow_v1_0`)
- `download_cg iar_tables()`: Download CGIAR-CSI tables (`cg iar_csi_v3`)
- `download_merit_dem_tables()`: Download MERIT-DEM tables (Multi-Error-Removed Improved-Terrain Digital Elevation Model, `merit_dem_v1_0_3`)
- `download_hydrography90m_tables()`: Download Hydrography90m tables (`hydrography90m_v1_0`)
- `download_observed_climate_tables()`: Download CHELSA bioclimatic variables tables, except for projections (Climatologies at high resolution for the earth's land surface areas, `chelsa_bioclim_v2_1`)
- `download_projected_climate_tables()`: Download CHELSA bioclimatic variables, tables, projections only (Climatologies at high resolution for the earth's land surface areas, `chelsa_bioclim_v2_1`)
- `download_landcover_tables()`: Download ESA Land Cover tables (`esa_cci_landcover_v2_1_1`)

Author(s)

Merret Buurman

References

Garcia Marquez J., Amatulli, G., Grigoropoulou, A., Schürz, M., Tomiczek, T., Buurman, M., Bremerich, V., Bego, K. and Domisch, S.: Global datasets of aggregated environmental variables at the sub-catchment scale for freshwater biodiversity modeling, in prep. Please contact the authors for more up-to-date citation info.

See Also

`[download_tiles()]` for downloading spatial layers (raster, vector) of the original Hydrography90m dataset, split to the same tiles.

Examples

```
### Soil: soilgrids250m_v2_0 ###
# Show all available soil variable names:
download_soil_tables()

# Compute download size of all soil variables, for one tile:
download_soil_tables(
  subset = "ALL",
  tile_ids = c("h00v04"),
  download = FALSE)
```



```

# Download one soil variable (Clay content), for two tiles:
## Not run:
vars <- download_soil_tables(
  subset = c("clyppt"),
  tile_ids = c("h00v04", "h10v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "zip")

## End(Not run)

# Download one soil variable (Clay content), for one tile,
# unzip, and delete the zips:
## Not run:
vars <- download_soil_tables(
  subset = c("clyppt"),
  tile_ids = c("h00v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "txt",
  delete_zips = TRUE)

## End(Not run)

### Flow (mean): flo1k_v1_0 ###
# Show all available flo1k variable names:
download_flo1k_tables()

# Compute download size of the only flo1k_v1_0 variable (mean flow),
# for one tile:
## Not run:
download_flo1k_tables(
  subset = "ALL",
  tile_ids = c("h00v04"),
  download = FALSE)

## End(Not run)

# Download the only flo1k_v1_0 variable (flo1k), for two tiles:
## Not run:
vars <- download_flo1k_tables(
  subset = c("flo1k"),
  tile_ids = c("h00v04", "h10v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "zip")

## End(Not run)

# Download the only flo1k_v1_0 variable (flo1k), for one tile,
# unzip, and delete the zips:

```

```

## Not run:
vars <- download_flo1k_tables(
  subset = c("flo1k"),
  tile_ids = c("h00v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "txt",
  delete_zips = TRUE)

## End(Not run)

### CGIAR CSI dataset: cgear_csi_v3 ###
# Show all available cgear variable names
download_cgear_tables()

# Compute download size of all cgear variables, for one tile:
download_cgear_tables(
  subset = "ALL",
  tile_ids = c("h00v04"),
  download = FALSE)

# Download one cgear variable (Global Aridity Index), for two tiles:
## Not run:
vars <- download_cgear_tables(
  subset = c("garid"),
  tile_ids = c("h00v04", "h10v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "zip")

## End(Not run)

# Download two cgear variables (Global Aridity Index, Potential
# Evapotranspiration), for one tile, unzip, and delete the zips:
## Not run:
vars <- download_cgear_tables(
  subset = c("garid", "gevapt"),
  tile_ids = c("h00v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "txt",
  delete_zips = TRUE)

## End(Not run)

### Digital Elevation Model: merit_dem_v1_0_3 ###
# (Multi-Error-Removed Improved-Terrain Digital Elevation Model)
# Show all available merit_dem_v1_0_3 variable names
download_merit_dem_tables()

# Compute download size of the only merit_dem_v1_0_3 variable

```

```

# (Mean elevation), for one tile:
download_merit_dem_tables(
  subset = "ALL",
  tile_ids = c("h00v04"),
  download = FALSE)

# Download the only merit_dem_v1_0_3 variable (mean elevation),
# for two tiles:
## Not run:
vars <- download_merit_dem_tables(
  subset = c("elev"), # or "ALL"
  tile_ids = c("h00v04", "h10v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "zip")

## End(Not run)

# Download the only merit_dem_v1_0_3 variable (mean elevation),
# for one tile, unzip, and delete the zips:
## Not run:
vars <- download_merit_dem_tables(
  subset = c("elev"),
  tile_ids = c("h00v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "txt",
  delete_zips = TRUE)

## End(Not run)

### Hydrography90m: hydrography90m_v1_0 ###
# Show all available hy90m variable names
download_hydrography90m_tables()

# Compute download size of all hy90m variables, for one tile:
download_hydrography90m_tables(
  subset = "ALL",
  tile_ids = c("h00v04"),
  download = FALSE)

# Download one hy90m variable (Strahler's stream order), for two tiles:
## Not run:
vars <- download_hydrography90m_tables(
  subset = c("order_strahler"),
  tile_ids = c("h00v04", "h10v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "zip")

## End(Not run)

```

```

# Download one hy90m variable (Strahler's stream order), for one tile,
# unzip, and delete the zips:
## Not run:
vars <- download_hydrography90m_tables(
  subset = c("order_strahler"),
  tile_ids = c("h00v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "txt",
  delete_zips = TRUE)

## End(Not run)

### Bioclimatic Variables: chelsa_bioclim_v2_1 ###
### (excluding projections) ###
# Show all available bioclim variable names
# (excluding projections):
download_observed_climate_tables()

# Compute download size of all bioclim variables, for one tile:
download_observed_climate_tables(
  subset = "ALL",
  tile_ids = c("h00v04"),
  download = FALSE)

# Download one bioclim variable (Annual mean temperature), for two tiles:
## Not run:
vars <- download_observed_climate_tables(
  subset = c("bio01_1981-2010_observed"),
  tile_ids = c("h00v04", "h10v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "zip")

## End(Not run)

# Download one bioclim variable (Annual mean temperature), for one tile,
# unzip, and delete the zips:
## Not run:
vars <- download_observed_climate_tables(
  subset = c("bio01_1981-2010_observed"),
  tile_ids = c("h00v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "txt",
  delete_zips = TRUE)

## End(Not run)

### Bioclimatic Variables: chelsa_bioclim_v2_1 ###
### (projections only) ###
# Show all available projected bioclim variable names

```

```

download_projected_climate_tables()

# Compute download size of all variables, for one tile:
download_projected_climate_tables(
  subset = "ALL",
  tile_ids = c("h00v04"),
  download = FALSE)

# Download one variable for two tiles:
## Not run:
vars <- download_projected_climate_tables(
  subset = c("bio01_2071-2100_ukesm1-0-ll_ssp585_v2_1"),
  tile_ids = c("h00v04", "h10v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "zip")

## End(Not run)

# Download one variable for two tiles, by specifying each part of
# the variable separately (e.g. scenario, model, time period, ...):
## Not run:
vars <- download_projected_climate_tables(
  base_vars = c("bio01"),
  time_periods = c("2071-2100"),
  models=c("ukesm1-0-ll"),
  scenarios=c("ssp585"),
  version=c("v2_1"),
  tile_ids = c("h00v04", "h10v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "zip")

## End(Not run)

# Download one variable for one tile,
# unzip, and delete the zips:
## Not run:
vars <- download_projected_climate_tables(
  subset = c("bio01_2071-2100_ukesm1-0-ll_ssp585_v2_1"),
  tile_ids = c("h00v04"),
  download = TRUE,
  download_dir = ".",
  file_format = "txt",
  delete_zips = TRUE)

## End(Not run)

### Landcover: esa_cci_landcover_v2_1_1 ###
# Show all available landcover variable names:
download_landcover_tables()

```

```
# Compute download size of two landcover base variables (Cropland, rainfed,
# and Grassland) and two years, for all tiles:
vars <- download_landcover_tables(
  base_vars=c("c10", "c200"),
  years=c(1992, 1993),
  tile_ids="ALL")

# Download two base variables and one year, for two tiles:
## Not run:
vars <- download_landcover_tables(
  base_vars=c("c10", "c200"),
  years=c(1992),
  tile_ids=c("h00v04", "h10v04"),
  download=TRUE,
  download_dir="/tmp",
  file_format="zip",
  delete_zips=FALSE)

## End(Not run)
```

download_test_data	<i>Download test data</i>
--------------------	---------------------------

Description

Download the test data of the package, which includes all Hydrography90m and species point observation data for a small geographic extent, to test the functions.

The test data will be automatically downloaded and unzipped with this function to a desired path, or can be alternatively downloaded at

https://drive.google.com/file/d/1kYNWxmtVm6X7MZLIS0ePGpvxB1pk1scD/view?usp=share_link.

Usage

```
download_test_data(download_dir = ".")
```

Arguments

download_dir character. The directory where the files will be downloaded. Default location is the working directory.

Details

Downloads the test data of the Hydrography90m dataset

Author(s)

Afroditi Grigoropoulou

References

Amatulli, G., Garcia Marquez, J., Sethi, T., Kiesel, J., Grigoropoulou, A., Üblacker, M. M., Shen, L. Q., and Domisch, S.: Hydrography90m: a new high-resolution global hydrographic dataset, Earth Syst. Sci. Data, 14, 4525–4550, <https://doi.org/10.5194/essd-14-4525-2022>, 2022.

Amatulli G., Garcia Marquez J., Sethi T., Kiesel J., Grigoropoulou A., Üblacker M., Shen L. & Domisch S. (2022-08-09). Hydrography90m: A new high-resolution global hydrographic dataset. IGB Leibniz-Institute of Freshwater Ecology and Inland Fisheries. dataset. <https://doi.org/10.18728/igb-fred-762.1>

Examples

```
# Download the test data to the current working directory
download_test_data()

# Download the data to a specific (existing) directory
download_test_data("path/to/your/directory")
```

download_tiles	<i>Download files of the Hydrography90m dataset</i>
----------------	---

Description

The function downloads data of the Hydrography90m dataset, which is split into 20°x20° tiles. If a tile ID is specified, then the selected layers (variable) will be downloaded. In addition, the Hydrography90m is organized in non-interrupted drainage basins called regional units. If a regional unit ID (reg_unit_id) is specified, then only the raster mask of the drainage basin is downloaded (useful for later processing). Multiple regular tiles, e.g. belonging to regional units, can be downloaded in a single request. The tile or regional unit IDs can be obtained using the functions "get_tile_id" and "get_regional_unit_id", respectively. The files will be stored locally in a folder architecture, similar as in the data repository, available at <https://public.igb-berlin.de/index.php/s/agciopgzXjWswF4?path=%2F>.

Usage

```
download_tiles(
  variable,
  file_format = "tif",
  tile_id = NULL,
  reg_unit_id = NULL,
  global = FALSE,
  download_dir = "."
)
```

Arguments

variable	character vector of variable names. See Details for all the variable names.
file_format	character. Format of the requested file ("tif" or "gpkg"). See Details.
tile_id	character vector. The IDs of the requested tiles.
reg_unit_id	character vector. The IDs of the requested regional units.
global	logical. If TRUE, the global extent file is downloaded. Default is FALSE.
download_dir	character. The directory where the files will be downloaded. Default is the working directory.

Details

In the following table you can find all the variables included in the Hydrography90m dataset. The column "Variable" includes the variable names that should be used as an input in the parameter "variable" of the function. Likewise, the column "File format" contains the input that should be given to the "file_format" parameter. For more details and visualisations of the spatial layers, please refer to https://hydrography.org/hydrography90m/hydrography90m_layers/.

Variable type	Variable name	Variable	Unit	File format
Network	Drainage basin	basin		tif
Network	Drainage basin	basin		gpkg
Network	Sub-catchment	sub_catchment		tif
Network	Sub-catchment	sub_catchment		gpkg
Network	Stream segment	segment		tif
Network	Outlet	outlet		tif
Network	Outlet	outlet		gpkg
Network	Regional unit	regional_unit		tif
Network	Flow direction	direction		tif
Flow	Flow accumulation	accumulation	km ²	tif
Stream slope	Cell maximum curvature	slope_curv_max_dw_cel	1/m	tif
Stream slope	Cell minimum curvature	slope_curv_min_dw_cel	1/m	tif
Stream slope	Cell elevation difference	slope_elv_dw_cel	m	tif
Stream slope	Cell gradient	slope_grad_dw_cel		tif
Stream distance	Shortest distance to drainage divide	stream_dist_up_near	m	tif
Stream distance	Longest distance to drainage divide	stream_dist_up_farth	m	tif
Stream distance	Nearest down stream stream grid cell	stream_dist_dw_near	m	tif
Stream distance	Outlet grid cell in the network	outlet_dist_dw_basin	m	tif
Stream distance	Down stream stream node grid cell	outlet_dist_dw_scatch	m	tif
Stream distance	Euclidean distance	stream_dist_proximity	m	tif
Elevation difference	Shortest path	stream_diff_up_near	m	tif
Elevation difference	Longest path	stream_diff_up_farth	m	tif
Elevation difference	Nearest downstream stream pixel	stream_diff_dw_near	m	tif
Elevation difference	Outlet grid cell in the network	outlet_diff_dw_basin	m	tif
Elevation difference	Downstream stream node grid cell	outlet_diff_dw_scatch	m	tif
Segment properties	Segment downstream mean gradient	channel_grad_dw_seg		tif
Segment properties	Segment upstream mean gradient	channel_grad_up_seg		tif
Segment properties	Cell upstream gradient	channel_grad_up_cel		tif

Segment properties	Cell stream course curvature	channel_curv_cel		tif
Segment properties	Segment downstream elevation difference	channel_elv_dw_seg		tif
Segment properties	Segment upstream elevation difference	channel_elv_up_seg		tif
Segment properties	Cell upstream elevation difference	channel_elv_up_cel		tif
Segment properties	Cell downstream elevation difference	channel_elv_dw_cel		tif
Segment properties	Segment downstream distance	channel_dist_dw_seg		tif
Segment properties	Segment upstream distance	channel_dist_up_seg		tif
Segment properties	Cell upstream distance	channel_dist_up_cel		tif
Stream order	Strahler's stream order	order_strahler		tif
Stream order	Shreve's stream magnitude	order_shreve		tif
Stream order	Horton's stream order	order_horton		tif
Stream order	Hack's stream order	order_hack		tif
Stream order	Topological dimension of streams	order_topo		tif
Stream order	Strahler's stream order	order_vect_segment		gpkg
Stream order	Shreve's stream magnitude	order_vect_segment		gpkg
Stream order	Horton's stream order	order_vect_segment		gpkg
Stream order	Hack's stream order	order_vect_segment		gpkg
Stream order	Topological dimension of streams	order_vect_segment		gpkg
Stream reach	Length of the stream reach	order_vect_segment	m	gpkg
Stream reach	Straight length	order_vect_segment	m	gpkg
Stream reach	Sinusoid of the stream reach	order_vect_segment		gpkg
Stream reach	Accumulated length	order_vect_segment	m	gpkg
Stream reach	Flow accumulation	order_vect_segment	km ²	gpkg
Stream reach	Distance to outlet	order_vect_segment	m	gpkg
Stream reach	Source elevation	order_vect_segment	m	gpkg
Stream reach	Outlet elevation	order_vect_segment	m	gpkg
Stream reach	Elevation drop	order_vect_segment		gpkg
Stream reach	Outlet drop	order_vect_segment		gpkg
Stream reach	Gradient	order_vect_segment		gpkg
Flow index	Stream power index	spi		tif
Flow index	Sediment transportation index	sti		tif
Flow index	Compound topographic index	cti		tif

Note

If there is an error during the download of a file (more likely in case of files bigger than 3-4GB), you can try to manually download this file by pasting the link that is returned by the error message in your browser.

Author(s)

Afroditi Grigoropoulou

References

Amatulli G., Garcia Marquez J., Sethi T., Kiesel J., Grigoropoulou A., Üblacker M., Shen L. & Domisch S. (2022-08-09) Hydrography90m: A new high-resolution global hydrographic dataset.

IGB Leibniz-Institute of Freshwater Ecology and Inland Fisheries. dataset. <https://doi.org/10.18728/igb-fred-762.1>

Examples

```
# Download data for two variables in three regular tiles
# to the current working directory
download_tiles(variable = c("sti", "stream_dist_up_farth"),
              file_format = "tif",
              tile_id = c("h00v02", "h16v02", "h16v04"))

# Download the global .tif layer for the variable "direction"
# into the temporary R folder or define a different directory
# Define directory
my_directory <- tempdir()
# Download layer
download_tiles(variable = "direction",
              file_format = "tif",
              global = TRUE,
              download_dir = my_directory)

# Download the raster mask of two regional units
# to the current working directory.
download_tiles(variable = "regional_unit",
              file_format = "tif",
              reg_unit_id = c("33", "34"))

# Download the raster mask of all regional units
# to the current working directory.
download_tiles(variable = "regional_unit",
              file_format = "tif",
              global = TRUE)
```

extract_from_gpkg	<i>Extract values from the stream order .gpkg files.</i>
-------------------	--

Description

The function reads the attribute table of the stream network GeoPackage file (.gpkg) stored on disk and extracts the data for one or more (or all) input sub-catchment (i.e. stream segment) IDs. The output is a data.table, and only the output is loaded into R.

Usage

```
extract_from_gpkg(
  data_dir,
  subc_id,
  subc_layer,
  var_layer,
```

```

    out_dir = NULL,
    file_name = NULL,
    n_cores = NULL,
    quiet = TRUE
)

```

Arguments

data_dir	character. Path to the directory containing all input data.
subc_id	a numeric vector of sub-catchment IDs or "all". If "all", the attribute table is extracted for all the stream segments of the input .gpkg layer. The stream segment IDs are the same as the sub-catchment IDs. A vector of the sub-catchment IDs can be acquired from the extract_ids() function, by sub-setting the resulting data.frame.
subc_layer	character. Full path to the sub-catchment ID .tif layer
var_layer	character vector of .gpkg files on disk, e.g. "order_vect_point_h18v04.gpkg".
out_dir	character. The directory where the output will be stored. If the out_dir is specified, the attribute tables will be stored as .csv files in this location, named after their input variable vector files (e.g. "/path/to/stats_order_vect_point_h18v04.csv"). If NULL, the output is only loaded in R and not stored on disk.
file_name	character. Name of the .csv file where the output table will be stored. out_dir should also be specified for this purpose.
n_cores	numeric. Number of cores used for parallelization, in case multiple .gpkg files are provided to var_layer. If NULL, available cores - 1 will be used.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Details

The following attributes are stored in the stream network .gpkg files (as produced by the GRASS GIS function r.stream.order:

- cat - category
- stream - sub-catchment / stream segment ID (equal to cat)
- next_stream - downstream sub-catchment / stream segment ID
- prev_streams; two or more upstream sub-catchment / stream segment IDs
- strahler - Strahler's stream order
- horton - Hortons's stream order
- shreve - Shreve's stream magnitude
- hack - Hack's main streams or Gravelius order
- topo_dim - Topological dimension streams order
- scheidegger - Scheidegger's Consisted Associated Integers
- drwal - Drwal's stream hierarchy
- length - length of the stream segment

- stright - length of the stream segment as a stright line
- sinusoid - fractal dimension: stream segment length / stright stream
- segment length;
- cum_length - length of the stream from the source
- flow_accum - flow accumulation within the sub-catchment of a stream segment
- out_dist - distance of current stream initialisation from outlet
- source_elev - elevation at stream segment initialisation
- outlet_elev - elevation at stream segment outlet
- elev_drop difference between source_elev and outlet_elev + drop outlet
- out_drop - drop at the outlet of the stream segment
- gradient - drop/length

Author(s)

Afroditi Grigoropoulou, Jaime Garcia Marquez, Marlene Schürz

References

<https://grass.osgeo.org/grass82/manuals/v.in.ogr.html> <https://grass.osgeo.org/grass82/manuals/addons/r.stream.order.html>

Examples

```
# Download test data into temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Define path to the directory containing all input data
test_data <- paste0(my_directory, "/hydrography90m_test_data")

# Define sub-catchment ID layer
subc_raster <- paste0(my_directory, "/hydrography90m_test_data",
                      "/subcatchment_1264942.tif")

# Extract the attribute table of the file order_vect_59.gpkg for all the
# sub-catchment IDs of the subcatchment_1264942.tif raster layer
attribute_table <- extract_from_gpkg(data_dir = test_data,
                                     subc_id = "all",
                                     subc_layer = subc_raster,
                                     var_layer = "order_vect_59.gpkg",
                                     n_cores = 1)

# Show the output table
attribute_table
```

extract_ids

*Extract sub-catchment and/or basin IDs***Description**

Extracts the ID value of the basin and/or sub-catchment raster layer at given point locations. Can also be used for point-based extraction of any .tif layer by specifying the layer in the "basin" parameter. The function can be used to extract all the sub-catchment IDs present in a sub-catchment raster as well, by providing the the sub-catchment raster layer as the only input.

Usage

```
extract_ids(
  data = NULL,
  lon,
  lat,
  id = NULL,
  basin_layer = NULL,
  subc_layer = NULL,
  quiet = TRUE
)
```

Arguments

data	a data.frame or data.table that contains the columns regarding the longitude / latitude coordinates in WGS84. If data is missing and a sub-catchment raster layer is provided, the function will extract all the ID value present in the layer.
lon	character. The name of the column with the longitude coordinates.
lat	character. The name of the column with the latitude coordinates.
id	character. The name of a column containing unique IDs for each row of "data" (e.g., occurrence or site IDs).
basin_layer	character. Full path to the .tif layer with the basin ID.
subc_layer	character. Full path to the .tif layer with the sub-catchment ID.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Details

For the extraction of a value at a given point location from the basin and/or sub-catchment raster layer of the Hydrography90m dataset, the GDAL function 'gdallocationinfo' is used. The point locations have to be defined by coordinates in the WGS84 reference system. The function can also be used to extract any value from a given raster layer in a WGS84 projection, such as environmental information that is stored in the input raster file. The extraction of all ID values from a sub-catchment raster layer is done with terra::unique.

Note

Duplicated rows will be removed.

Author(s)

Afroditi Grigoropoulou, Marlene Schürz

References

<https://gdal.org/programs/gdallocationinfo.html>

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Example 1: Extracts the ID value at given point locations

# Load occurrence data
species_occurrence <- read.table(paste0(my_directory,
                                         "/hydrography90m_test_data",
                                         "/spdata_1264942.txt"),
                                header = TRUE)

# Define full path to the basin and sub-catchments raster layer
basin_raster <- paste0(my_directory,
                      "/hydrography90m_test_data/basin_1264942.tif")
subc_raster <- paste0(my_directory,
                     "/hydrography90m_test_data/basin_1264942.tif")

# Extract basin and sub-catchment IDs from the Hydrography90m layers
hydrography90m_ids <- extract_ids(data = species_occurrence,
                                lon = "longitude",
                                lat = "latitude",
                                id = "occurrence_id",
                                subc_layer = subc_raster,
                                basin_layer = basin_raster)

# Show the output table
hydrography90m_ids

# Example 2: Extract ID values of all subcatchments
subc_raster <- paste0(my_directory,
                     "/hydrography90m_test_data/subcatchment_1264942.tif")
hydrography90m_ids <- extract_ids(subc_layer = subc_raster)
fwrite(hydrography90m_ids, paste0(my_directory, '/subc_IDs.txt'))
```

extract_lake_ids	<i>Extract lake ids</i>
------------------	-------------------------

Description

Extract lake ids either at the location of point occurrences or falling within a bounding box

Usage

```
extract_lake_ids(
  data,
  lon,
  lat,
  lake_shape,
  var_name = "Hylak_id",
  xmin = NULL,
  ymin = NULL,
  xmax = NULL,
  ymax = NULL,
  bbox = FALSE,
  lake_id_table,
  quiet = TRUE
)
```

Arguments

data	a data.frame or data.table that contains the columns the longitude / latitude coordinates in WGS84, i.e. of species point occurrences
lon	character. The name of the column with the longitude coordinates.
lat	character. The name of the column with the latitude coordinates.
lake_shape	Full path to lake shape files; i.e. HydroLAKES shapefiles; when NULL a lake raster file needs to be provided
var_name	character. Name of the shape file ID column, i.e. "Hylak_id" for HydroLAKES; default is "Hylak_id"
xmin	integer. bounding box coordinate check what is lower left corner in WGS84; if NULL see parameter bbox
ymin	integer. bounding box coordinate check what is lower right corner in WGS84; if NULL see parameter bbox
xmax	integer. bounding box coordinate upper right corner in WGS84; if NULL see parameter bbox
ymax	integer. bounding box coordinate upper left corner in WGS84; if NULL see parameter bbox

bbox	logical. if TRUE, then all lake ids within the bounding box are returned, note that either bounding box values can be supplied thorough xmin, ymin, xmax and ymax values or when left empty (NULL) bounding box is calculated from min max values of the longitude and latitude values provided in the occurrence table; if FALSE lake ids in which point occurrences are located are returned, default is FALSE
lake_id_table	character. Full path of the output lake id table
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Details

For the extraction of a value at a given point location from the basin and/or sub-catchment raster layer of the Hydrography90m dataset, the GDAL function 'gdallocationinfo' is used. The point locations have to be defined by coordinates in the WGS84 reference system.

Author(s)

Jaime Garcia Marquez, Thomas Tomiczek

References

<https://grass.osgeo.org/grass82/manuals/> <https://gdal.org/programs/gdallocationinfo.html>

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Example 1: Extracts lake IDs for points located within a lake

# Load occurrence data

species_occurrence <- read.table(paste0(my_directory,
                                          "/hydrography90m_test_data",
                                          "/spdata_1264942.txt"),
                                 header = TRUE)

# extract lake IDs which intersect with species point occurrences
extract_lake_ids(data = species_occurrence,
                 lon = "longitude",
                 lat = "latitude",
                 var_name = "lake_id",
                 lake_shape = paste0(my_directory, "/hydrography90m_test_data",
                                     "/hydrography90m_test_lakes.gpkg"),
                 lake_id_table = output_folder,
                 quiet = FALSE)

# extract lake IDs from bounding box taken from species point occurrences
extract_lake_ids(data = species_occurrence,
```



```

lon = "longitude",
lat = "latitude",
var_name = "lake_id",
lake_shape = paste0(my_directory, "/hydrography90m_test_data",
                    "/hydrography90m_test_lakes.gpkg"),
bbox = TRUE,
lake_id_table = output_folder,
quiet = FALSE)

# extract lake IDs from provided bounding box
extract_lake_ids(data = species_occurrence,
lon = "longitude",
lat = "latitude",
xmin = 9.498263,
ymin = 44.21651,
xmax = 29.71806,
ymax = 50.25584,
var_name = "lake_id",
lake_shape = paste0(my_directory, "/hydrography90m_test_data",
                    "/hydrography90m_test_lakes.gpkg"),
bbox = TRUE,
lake_id_table = output_folder,
quiet = FALSE)

```

extract_partial_stream_network

Extract partial stream network needed for snapped points

Description

Given a stream network and snapped point subcatchments, this returns the reduced stream network consisting of: - All segments with Strahler order $\geq$ 'strahler_retain_threshold' - All downstream paths from finer-order snapped segments until reaching a segment with Strahler $\geq$ 'strahler_retain_threshold' - PLUS a configurable number of *upstream* same-Strahler segments for context

Usage

```

extract_partial_stream_network(
  stream,
  snapped_subcs,
  strahler_retain_threshold = 4,
  upstream_buffer = 0
)

```

Arguments

stream	sf object with columns: 'subc_id', 'target', 'strahler_order', and LINESTRING geometry.
snapped_subcs	integer vector: subc_ids after snapping.

strahler_retain_threshold
integer: minimum Strahler to fully retain (default 4).

upstream_buffer
integer: how many same-Strahler upstream segments to retain.

Value

sf object containing the partial stream network.

extract_zonal_stat	<i>Calculate zonal statistics</i>
--------------------	-----------------------------------

Description

Calculate zonal statistics based on one or more environmental variable raster .tif layers. This function aggregates data to 12 summary statistics (mean, min, max, range, ...) for selected or all sub-catchments of the input file. The sub-catchment raster (.tif) input file is read directly from disk. The output is a data.table which is loaded into R. This function can also be used for any zonal statistic calculation by specifying the raster layer zones in the subc_layer parameter and optionally, also the target zone IDs in the subc_id parameter.

Usage

```
extract_zonal_stat(
  data_dir,
  subc_id,
  subc_layer,
  var_layer,
  out_dir = NULL,
  file_name = NULL,
  n_cores = NULL,
  quiet = TRUE
)
```

Arguments

data_dir	character. Path to the directory containing all input data.
subc_id	Vector of sub-catchment IDs or "all". If "all", zonal statistics are calculated for all sub-catchments of the given sub-catchment raster layer. A vector of the sub-catchment IDs can be acquired from the extract_ids() function, and by sub-setting the resulting data.frame.
subc_layer	character. Full path to the sub-catchment ID .tif layer.
var_layer	character vector of variable raster layers on disk, e.g. "slope_grad_dw_cel_h00v00.tif". Note that the variable name appears in the output table columns (e.g. slope_grad_dw_cel_mean). To speed up the processing, the selected variable raster layers can be cropped to the extent of the sub-catchment layer, e.g. with crop_to_extent() .

out_dir	character. The directory where the output will be stored. If the out_dir and file_name are specified, the output table will be stored as a .csv file in this location. If they are NULL, the output is only loaded in R and not stored on disk.
file_name	character. Name of the .csv file where the output table will be stored. out_dir should also be specified for this purpose.
n_cores	numeric. Number of cores used for parallelization, in case multiple .tif files are provided to var_layer. Default is 1.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Value

Returns a table with

- sub-catchment ID (subc_id)
- number of cells with a value (data_cells)
- number of cells with a NoData value (nodata_cells)
- minimum value (min)
- maximum value (max)
- value range (range)
- arithmetic mean (mean)
- arithmetic mean of the absolute values (mean_abs)
- standard deviation (sd)
- variance (var)
- coefficient of variation (cv)
- sum (sum)
- sum of the absolute values (sum_abs).

Author(s)

Afroditi Grigoropoulou, Jaime Garcia Marquez, Marlene Schürz

References

<https://grass.osgeo.org/grass82/manuals/r.univar.html>

See Also

- [report_no_data\(\)](#) to check the defined NoData value.
- [set_no_data\(\)](#) to define a NoData value.
- [crop_to_extent\(\)](#) to crop the data to the same extent as the sub-catchments (subc_layer).

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Define full path to the sub-catchment ID .tif layer
subc_raster <- paste0(my_directory, "/hydrography90m_test_data",
                      "/subcatchment_1264942.tif")

# Define the directory where the output will be stored
output_folder <- paste0(my_directory, "/hydrography90m_test_data/output")
# Create output folder if it doesn't exist
if(!dir.exists(output_folder)) dir.create(output_folder)

# Calculate the zonal statistics for all sub-catchments for two variables
stat <- extract_zonal_stat(data_dir = paste0(my_directory,
                                              "/hydrography90m_test_data"),
                          subc_id = c(513837216, 513841103,
                                       513850467, 513868394,
                                       513870312),
                          subc_layer = subc_raster,
                          var_layer = c("spi_1264942.tif",
                                         "sti_1264942.tif"),
                          out_dir = output_folder,
                          file_name = "zonal_statistics.csv",
                          n_cores = 2)

# Show output table
stat
```

```
get_all_upstream_distances
```

Get all upstream distances for all subc_id

Description

Calculates the upstream distance from each subc_id to all upstream subc_id. The output can be directly used in spatial prioritization analyses for e.g. Marxan, Gurobi etc. to specify the longitudinal connectivity. Note that the stream segment and sub-catchment IDs are identical, and for consistency, we use the term "subc_id".

Note that the distance can be extremely long and for the subsequent spatial prioritization analyses you might want to consider setting a cap at a certain distance.

Usage

```
get_all_upstream_distances(network_table = network_table, n_cores = 1)
```

Arguments

- network_table** A data.table that includes the columns c(stream, next_stream, out_dist, the latter specifying the distance to the outlet (which is included in the Hydrography90m vector attribute table).
- n_cores** numeric. Number of cores used for parallelisation in the case of multiple stream segments / s. Default is 1. Currently, the parallelisation process requires copying the data to each core. In case the graph is very large, and many segments are used as an input, setting n_cores to a higher value can speed up the computation. This comes however at the cost of possible RAM limitations and even slower processing since the large data will be copied to each core. Hence consider testing with n_cores = 1 first. Optional.

Value

A data.table that reports the distance (in meters) from each subc_id to all upstream subc_ids.

Note

Currently the attributes are not provided for the (the selected subc_id segment). If the attributes are also needed for the outlet subc_id, then the next downstream sub_id can be selected (enlarge the study area)

Author(s)

Sami Domisch

References

Csardi G, Nepusz T: The igraph software package for complex network research, InterJournal, Complex Systems 1695. 2006. <https://igraph.org>

See Also

[read_geopackage\(\)](#) and [get_catchment_graph\(\)](#) to create a network graph. Alternatively, see [get_segment_neighbours\(\)](#) to obtain the upstream variables for a specified neighbourhood, or [get_upstream_variable\(\)](#) to aggregate a set of variables across the upstream catchment.

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Load stream network as a graph
my_graph <- read_geopackage(gpkg = paste0(my_directory,
                                           "/hydrography90m_test_data",
                                           "/order_vect_59.gpkg"),
                           import_as = "graph")
```

```

# Pick a random subc_id
subc_id = "513867228"
# Get the upstream catchment as a data.table
network_table <- hydrographr::get_catchment_graph(g = my_graph ,
                                                  subc_id = subc_id,
                                                  mode = "in",
                                                  use_outlet = FALSE,
                                                  as_graph = FALSE,
                                                  n_cores = 1)

## Condense the table supplied to the function to save RAM
keep_these <- c("stream", "next_stream", "out_dist")
network_table <- network_table[, ..keep_these]

## Change to integers
network_table$stream <- as.integer(network_table$stream)
network_table$next_stream <- as.integer(network_table$next_stream)

## Calculate the network distance (in meter) from each subc_id to
## all upstream subc_id using four CPUs for the parallelization
result <- get_all_upstream_distances(network_table = network_table,
                                     n_cores = 4)

```

get_buffer_along_the_network

Get upstream and downstream network buffer along stream segments

Description

Given a stream network and starting points (either segment IDs or sf points), this function extracts the segments along the network within a specified upstream and/or downstream distance (buffer). It handles points snapped to stream segments and can clip partial segments based on the start fraction.

Usage

```

get_buffer_along_the_network(
  lines_sf,
  target_ids = NULL,
  start_points = NULL,
  up_radius = 0,
  down_radius = 0
)

```

Arguments

lines_sf	An 'sf' object representing the stream network. Must have columns 'subc_id' (segment ID), 'target' (downstream segment), 'length' (segment length), and LINESTRING geometry.
----------	--

target_ids	Optional integer or character vector of segment IDs to start the buffer from. Provide either 'target_ids' OR 'start_points', not both.
start_points	Optional 'sf' object of POINT geometries representing starting locations on the network. Provide either 'target_ids' OR 'start_points'.
up_radius	Numeric. Distance (in units of 'lines_sf') to trace upstream. Default is 0 (no upstream buffer).
down_radius	Numeric. Distance (in units of 'lines_sf') to trace downstream. Default is 0 (no downstream buffer).

Details

- Supports both single segments ('target_ids') or arbitrary points ('start_points') - Clips segments proportionally if the buffer does not reach the full segment length - Handles branching upstream segments and avoids duplicate segments for the same point - Requires 'sf', 'dplyr', and helper functions 'traverse_upstream()' and 'traverse_downstream()'

Value

An 'sf' object containing all segments within the requested upstream and downstream distances for each starting point. Additional columns: - 'point_id': identifier of starting point - 'target_segment': ID of segment used as starting point - 'start_fraction': fraction along starting segment where the point lies - 'direction': "upstream" or "downstream" - 'search_distance': the requested buffer distance

get_catchment_graph	<i>Get catchment from stream network graph</i>
---------------------	--

Description

Subset the stream network graph by extracting the upstream, downstream or entire catchment, for one or multiple stream segments. The function will return either one or more data.tables or graph objects for each input stream segment. Note that the stream segment and sub-catchment IDs are identical, and for consistency, we use the term "subc_id".

By switching the mode to either "in", "out" or "all", only the upstream, downstream or all connected segments will be returned, respectively. The function [read_geopackage\(\)](#) can be used to create the input network graph.

Usage

```
get_catchment_graph(
  g,
  subc_id = NULL,
  use_outlet = FALSE,
  mode = NULL,
  as_graph = FALSE,
  n_cores = 1,
  max_size = 1500
)
```

Arguments

<code>g</code>	igraph object. A directed graph.
<code>subc_id</code>	numeric vector of a single or multiple IDs, e.g (c(ID1, ID2, ID3, ...)). The sub-catchment (equivalent to stream segment) IDs for which to delineate the upstream drainage area. If empty, then outlets will be used as sub-catchment IDs (with <code>use_outlet = TRUE</code>). Note that you can browse the entire network online at https://geo.igb-berlin.de/maps/351/view and to left hand side, select the "Stream segment ID" layer and click on the map to get the ID. Optional.
<code>use_outlet</code>	logical. If TRUE, the outlets of the given network graph will be used as additional input <code>subc_ids</code> . Outlets will be identified internally as those stream segments that do not have any downstream connected segment. Default is FALSE.
<code>mode</code>	character. One of "in", "out" or "all". "in" returns the upstream catchment, "out" returns the downstream catchment (all catchments that are reachable from the given input segment), and "all" returns both.
<code>as_graph</code>	logical. If TRUE, the output will be a new graph or a list of new graphs with the original attributes. If FALSE, the output will be a new data.table or a list of data.tables. List objects are named after the <code>subc_ids</code> . Default is FALSE.
<code>n_cores</code>	numeric. Number of cores used for parallelisation in the case of multiple stream segments / s. Default is 1. Currently, the parallelisation process requires copying the data to each core. In case the graph is very large, and many segments are used as an input, setting <code>n_cores</code> to a higher value can speed up the computation. This comes however at the cost of possible RAM limitations and even slower processing since the large data will be copied to each core. Hence consider testing with <code>n_cores = 1</code> first. Optional.
<code>max_size</code>	numeric. Specifies the maximum size of the data passed to the parallel back-end in MB. Default is 1500 (1.5 GB). Consider a higher value for large study areas (more than one 20°x20° tile). Optional.

Value

A graph or data.table that reports all `subc_ids`. In case of multiple input segments, the results are stored in a list.

Note

Currently the attributes are not provided for the (the selected `subc_id` segment). If the attributes are also needed for the outlet `subc_id`, then the next downstream `sub_id` can be selected (enlarge the study area)

Author(s)

Sami Domisch

References

Csardi G, Nepusz T: The igraph software package for complex network research, InterJournal, Complex Systems 1695. 2006. <https://igraph.org>

See Also

[read_geopackage\(\)](#) to create a network graph.

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Load stream network as a graph
my_graph <- read_geopackage(gpkg = paste0(my_directory,
                                           "/hydrography90m_test_data",
                                           "/order_vect_59.gpkg"),
                           import_as = "graph")

# Pick a random subc_id
subc_id = "513855877"
# Get the upstream catchment as a graph
g_up <- get_catchment_graph(g = my_graph, subc_id = subc_id, mode = "in",
                           use_outlet = FALSE, as_graph = TRUE, n_cores = 1)

# Get the downstream segments as a data.table,
g_down <- get_catchment_graph(g = my_graph, subc_id = subc_id, mode = "out",
                           use_outlet = FALSE, as_graph = FALSE, n_cores = 1)

# Get the catchments of all outlets in the study area as a graph
g_all <- get_catchment_graph(g = my_graph, mode = "in", use_outlet = TRUE,
                           as_graph = TRUE, n_cores = 1)
```

get_centrality

Get centrality indexes from stream network graph

Description

Calculate centrality indexes from a directed stream network graph. By switching the mode to either "in", "out" or "all", only the upstream, downstream or all connected segments will be considered, respectively. The function [read_geopackage\(\)](#) can be used to create the input network graph.

Usage

```
get_centrality(g, index = "all", mode = NULL)
```

Arguments

g	igraph object. A directed graph.
index	character. One of "all", "closeness", "farness", "betweenness", "degree", "eccentricity". See @Details


```

                                "/order_vect_59.gpkg"),
import_as = "SpatVect")

# Merge the centrality table with the vector
stream_vect <- terra::merge(stream_vect, centrality,
                             by.x = c('stream'), by.y="subc_id")

# Write out the stream network vector including the centrality indices
writeVector(stream_vect, paste0(my_directory,
                                "/hydrography90m_test_data",
                                "/order_vect_59_centra.gpkg"))

```

get_distance	<i>Calculate euclidean or along the network distance between points located in one basin</i>
--------------	--

Description

Calculate euclidean or along-the-network distance (in meters) between points located in one basin. To calculate the distance along the network, point coordinates need to be snapped to the stream network using the function [snap_to_network\(\)](#) or [snap_to_subc_segment\(\)](#).

Usage

```

get_distance(
  data,
  lon,
  lat,
  id,
  stream_layer = NULL,
  distance = "both",
  n_cores = 1,
  quiet = TRUE
)

```

Arguments

data	a data.frame or data.table that contains the columns regarding the longitude / latitude coordinates in WGS84.
lon	character. The name of the column with the longitude coordinates.
lat	character. The name of the column with the latitude coordinates.
id	character. The name of a column containing unique IDs for each row of "data" (e.g., occurrence or site IDs). The unique IDs need to be numeric and less than 10 characters long.
stream_layer	character. Full path of the stream network .gpkg file. Needs to be defined to calculate the distance along the network. (In the case of Hydrography90m, the relevant files have the format "order_vect_segment_h??v??.gpkg")

distance	character. One of "euclidean", "network", or "both". If "euclidean", the euclidean distances between all pairs of points are calculated. If "network", the shortest path along the network between all pairs of points is calculated. (see "Details" for more information). If method is set to "both", both distance measures are calculated. Distances are given in meters. Default is "both".
n_cores	numeric. Number of cores used for parallelisation. Default is 1.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Details

To calculate the euclidean distance between all pairs of points the function uses the `v.distance` command of GRASS GIS, which has been set up to produce a square matrix of distances. The calculation of distances along the stream network has been implemented with the command `v.net.allpairs` of GRASS GIS. The along-the-network distance calculation is done for all pairs of points located within the same basin. If the points are located in different basins, the function `get_distance_parallel()` should be used.

Value

If `distance='euclidean'`, a distance matrix, in meters, of the euclidean distances between all the pairs of points (object of class `data.frame`) is returned. If `distance='network'`, a `data.frame` with three columns: `from_id`, `to_id`, `dist` is returned. The `'dist'` column includes the distance, in meters, of the shortest path along the network from the point `"from_id"` to the point `"to_id"`. If `distance='both'`, a list containing both objects is returned.

Author(s)

Afroditi Grigoropoulou, Marlene Schürz, Jaime Garcia Marquez

References

<https://grass.osgeo.org/grass82/manuals/v.net.allpairs.html> <https://grass.osgeo.org/grass82/manuals/v.distance.html>

See Also

- `snap_to_network()` to snap the data points to the next stream segment within a given radius and/or a given flow accumulation threshold value.
- `snap_to_subc_segment()` to snap the data points to the next stream segment of the sub-catchment the data point is located.
- `get_distance_parallel()` to calculate the distance along the network in more than two basins in parallel.

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)
```

```

# Load occurrence data
species_occurrence <- read.table(paste0(my_directory,
                                          "/hydrography90m_test_data/spdata_1264942.txt"),
                                header = TRUE)

basin_rast <- paste0(my_directory,
                    "/hydrography90m_test_data/basin_1264942.tif")

# Define full path to the sub-catchment raster layer
subc_rast <- paste0(my_directory,
                    "/hydrography90m_test_data/subcatchment_1264942.tif")

# Define full path to the vector file of the stream network
stream_vect <- paste0(my_directory,
                      "/hydrography90m_test_data/order_vect_59.gpkg")

# Automatically extract the basin and sub-catchment IDs and
# snap the data points to the stream segment
snapped_coordinates <- snap_to_subc_segment(data = species_occurrence,
                                           lon = "longitude",
                                           lat = "latitude",
                                           id = "occurrence_id",
                                           basin_layer = basin_rast,
                                           subc_layer = subc_rast,
                                           stream_layer = stream_vect,
                                           n_cores = 2)

# Show head of output table
head(snapped_coordinates)

# Get the euclidean distance and the distance along the network between all
# pairs of points
distance_table <- get_distance(data = snapped_coordinates,
                              lon = "lon_snap",
                              lat = "lat_snap",
                              id = "occurrence_id",
                              stream_layer = stream_vect,
                              distance = "network")

# Show table
distance_table

```

get_distance_graph	<i>Get the distance in meters between stream segments along a network graph</i>
--------------------	---

Description

Given a set of input sub-catchment IDs, the function will calculate the network distance in meters between all input pairs. Alternatively, only the number of stream segments (sub-catchment) along

the paths are reported. Note that the stream segment and sub-catchment IDs are identical, and for consistency, we use the term "subc_id".

See the example code under "Note" that explains how to use standard igraph functions to obtain the actual stream segment IDs along the path. The function `read_geopackage()` can be used to create the input network graph, and the function `get_catchment_graph()` can be used to subset a network graph. Note that graph-based function includes also the entire length of the "from" and "to" stream segment, opposed to the `get_distance()` which, depending on the snapping method, includes only the part of the "from" and "to" segment where the points are located, resulting in minor differences.

Usage

```
get_distance_graph(
  g,
  subc_id = NULL,
  variable = "length",
  distance_m = TRUE,
  max_size = 1500
)
```

Arguments

<code>g</code>	igraph object. A directed graph.
<code>subc_id</code>	numeric. Vector of a single or multiple IDs, e.g (c(ID1, ID2, ID3, ...)). The sub-catchment (equivalent to stream segment) IDs for calculating the distances. Note that you can browse the entire network online at https://geo.igb-berlin.de/maps/351/view and to the left hand side, select the "Stream segment ID" layer and click on the map to get the ID.
<code>variable</code>	character. Specify the attribute / column name in the graph object that should be cumulated along the network path. Default is "length" to be used with the Hydrography90m dataset. Not needed when using "distance_m = FALSE".
<code>distance_m</code>	logical. If TRUE, and in case of the Hydrography90m dataset, the length (in meters) of each network segment along the path will be cumulated and the total length between all pairs will be reported in data.table. If FALSE, only the number of segments that are traversed through will be reported in the output matrix. If the subc_ids of the actual path is needed, please see the example code under "Note". Default is TRUE.
<code>max_size</code>	numeric. Specifies the maximum size of the data passed to the parallel back-end in MB. Default is 1500 (1.5 GB). Consider a higher value for large study areas (more than one 20°x20° tile). Optional.

Value

A data.table that reports either the distance or the number of segments between sub_ids.


```

                                8.898060225276105),
latitude = c(42.26533822791161,
             42.26307509726402,
             42.25513157316764),
occurrence_id = c(1, 2, 3))

# Define the path of the sub-catchment raster
subc_raster <- paste0(my_directory,
                     "/hydrography90m_test_data/subcatchment_1264942.tif")

# Extract the sub-catchment IDs for the points:
my_points_subc_id <- extract_ids(data = my_points,
                                lon = "longitude",
                                lat = "latitude",
                                id = "occurrence_id",
                                subc_layer = subc_raster)

# Get a vector of the sub-catchment IDs:
subc_id <- my_points_subc_id$subcatchment_id

# Get the network distance (in meters) between all input pairs:
subc_distances <- get_distance_graph(my_graph,
                                    subc_id = subc_id,
                                    variable = "length",
                                    distance_m = TRUE)

subc_distances

# Get the number of stream segments that are along the network path:
number_segments <- get_distance_graph(my_graph,
                                     subc_id = subc_id,
                                     variable = "length",
                                     distance_m = FALSE)

number_segments

```

get_distance_parallel Calculate euclidean or along the network distance between points

Description

Calculate euclidean or along-the-network distance (in meters) between points. To calculate the distance along the network, point coordinates need to be snapped to the stream network using the function `snap_to_network()` or `snap_to_subc_segment()`.

Usage

```

get_distance_parallel(
  data,
  lon,

```



```

    lat,
    id,
    basin_id = NULL,
    basin_layer = NULL,
    stream_layer = NULL,
    distance = "both",
    n_cores = 1,
    quiet = TRUE
)

```

Arguments

data	a data.frame or data.table that contains the columns regarding the longitude / latitude coordinates in WGS84.
lon	character. The name of the column with the longitude coordinates.
lat	character. The name of the column with the latitude coordinates.
id	character. The name of a column containing unique IDs for each row of "data" (e.g., occurrence or site IDs). The unique IDs need to be numeric and less than 10 characters long.
basin_id	character. The name of the column with the basin IDs. If NULL and distance is set to 'network' or 'both', the basin IDs will be extracted automatically. Default is NULL.
basin_layer	character. Full path to the basin ID .tif layer. Needs to be defined to calculate the distance along the network.
stream_layer	character. Full path of the stream network .gpkg file. Needs to be defined to calculate the distance along the network.
distance	character. One of "euclidean", "network", or "both". If "euclidean", the euclidean distances between all pairs of points are calculated. If "network", the shortest path along the network between all pairs of points is calculated. (see "Details" for more information). If method is set to "both", both distance measures are calculated. Distances are given in meters. Default is "both".
n_cores	numeric. Number of cores used for parallelisation. Default is 1.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Details

To calculate the euclidian distance between all pairs of points the function uses the `v.distance` command of GRASS GIS, which has been set up to produce a square matrix of distances. The calculation of distances along the stream network has been implemented with the command `v.net.allpairs` of GRASS GIS. The along-the-network distance calculation is done for all pairs of points located within the same basin. If the points are located in different basins, the function can be run in parallel (i.e., each core for the distance calculations of all points within one basin). The distance between points located in different basins is zero because they are not connected through the network.

Value

If distance='euclidean', a distance matrix, in meters, of the euclidean distances between all the pairs of points (object of class data.frame) is returned. If distance='network', a data.frame with three columns: from_id, to_id, dist is returned. The 'dist' column includes the distance, in meters, of the shortest path along the network from the point "from_id" to the point "to_id". If distance='both', a list containing both objects is returned.

Author(s)

Afroditi Grigoropoulou, Marlene Schürz, Jaime Garcia Marquez

References

<https://grass.osgeo.org/grass82/manuals/v.net.allpairs.html> <https://grass.osgeo.org/grass82/manuals/v.distance.html>

See Also

- [get_distance\(\)](#) to calculate the distance along the network in points located in only one basin.
- [snap_to_network\(\)](#) to snap the data points to the next stream segment within a given radius and/or a given flow accumulation threshold value.
- [snap_to_subc_segment\(\)](#) to snap the data points to the next stream segment of the sub-catchment the data point is located.
- [extract_ids\(\)](#) to extract basin and sub-catchment IDs.

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Load occurrence data
species_occurrence <- read.table(paste0(my_directory,
                                          "/hydrography90m_test_data/spdata_1264942.txt"),
                                 header = TRUE)

basin_rast <- paste0(my_directory,
                    "/hydrography90m_test_data/basin_1264942.tif")

# Define full path to the sub-catchment raster layer
subc_rast <- paste0(my_directory,
                   "/hydrography90m_test_data/subcatchment_1264942.tif")

# Define full path to the vector file of the stream network
stream_vect <- paste0(my_directory,
                     "/hydrography90m_test_data/order_vect_59.gpkg")

# Automatically extract the basin and sub-catchment IDs and
```

```

# snap the data points to the stream segment
snapped_coordinates <- snap_to_subc_segment(data = species_occurrence,
                                           lon = "longitude",
                                           lat = "latitude",
                                           id = "occurrence_id",
                                           basin_layer = basin_rast,
                                           subc_layer = subc_rast,
                                           stream_layer = stream_vect,
                                           n_cores = 2)

# Show head of output table
head(snapped_coordinates)

# Get the euclidean distance and the distance along the network between all
# pairs of points
distance_table <- get_distance_parallel(data = snapped_coordinates,
                                       lon = "lon_snap",
                                       lat = "lat_snap",
                                       id = "occurrence_id",
                                       basin_id = "basin_id",
                                       basin_layer = basin_rast,
                                       stream_layer = stream_vect,
                                       distance = "network")

# Show table
distance_table

```

get_lake_catchment	<i>Get lake catchments based on intersection points between stream network and lakes</i>
--------------------	--

Description

Delineate lake catchments in GeoTIFF format based on intersection points between stream network (i.e. Hydrography90m) and lake geospatial files (e.g. HydroLAKES). The function uses the lake intersection tables from get_lake_intersection.R as input data.

Usage

```

get_lake_catchment(
  data,
  flow = "flow_accu_mean",
  lake_id = "lake_ID",
  n = "all",
  direction,
  lake_basin,
  n_cores = 1,
  quiet = TRUE
)

```

Arguments

<code>data</code>	a data.frame or data.table that contains the columns regarding the stream segment ids of the intersection points between lake and stream network; (i.e., output of <code>get_lake_intersection</code>)
<code>flow</code>	character. The name of the flow accumulation column used for sorting the intersection table; i.e. either <code>flow_accu</code> (flow accumulation value at intersection point pixel), <code>flow_accu_max</code> (maximum flow accumulation value of 3 x 3 neighboring pixels),
<code>lake_id</code>	character. The name of the column containing lake ids; (i.e., output "lake_ID" of <code>get_lake_intersection</code>) <code>flow_accu_mean</code> (mean flow accumulation value of 3 x 3 neighboring pixels). Default is <code>flow_accu_mean</code>
<code>n</code>	integer. Number of intersection points used for lake catchment delineation; e.g. <code>n=1</code> equals first row of intersection table and lake outlet; <code>n=20</code> equals first 20 rows of intersection table; Default is all.
<code>direction</code>	character. Full path to Hydrography90m flow direction tif file
<code>lake_basin</code>	character. Full path to output catchment tif files
<code>n_cores</code>	integer. Number of cores used in parallelization; Default is one.
<code>quiet</code>	logical. If FALSE, the standard output will be printed.; Default is TRUE.

Note

For the function to work we need the output of the function `get_lake_intersection` that are the `lake_referenceID.txt` and intersection table (i.e. `coord_lakeID.txt`)

Author(s)

Jaime Garcia Marquez, Thomas Tomiczek

References

add reference manual html here <https://grass.osgeo.org/grass82/manuals/>

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

data <- fread(paste0(my_directory,
                     "/hydrography90m_test_data",
                     "/coord_lake_1.txt"),
             header = TRUE)

direction <- (paste0(my_directory,
                     "/hydrography90m_test_data",
                     "/direction_1264942.tif"))
```

```
catch <- (paste0(my_directory,
                 "/hydrography90m_test_data/"))

get_lake_catchment(data, direction = direction, lake_basin = catch)
```

`get_lake_intersection` *Get intersection points between Hydrography90m stream network and lake shapefiles*

Description

The function identifies, for each lake ID, the intersection points between the stream network (i.e., Hydrography90m) and the corresponding lake's geospatial file (e.g., HydroLAKES shapefile).

Usage

```
get_lake_intersection(
  data,
  lake_id = "HydroLAKES_polys_v10",
  lakes,
  lake_name,
  buffer = TRUE,
  edge,
  stream,
  flow,
  basins,
  lake_dat,
  n_cores = 1,
  quiet = TRUE
)
```

Arguments

<code>data</code>	a data.frame or data.table that contains the columns regarding the lake ids, i.e. HydroLAKES ids. (i.e., output of <code>extract_lake_ids</code>); see also <code>help(extract_lake_ids)</code>
<code>lake_id</code>	character. The name of the column containing lake ids.
<code>lakes</code>	character. Full path to lake geo-spatial files; i.e. HydroLAKES shapefiles.
<code>lake_name</code>	character. The name of the attribute table column in the geo-spatial files containing the lake ids; i.e. "HydroLAKES_polys_v10" for HydroLAKES shapefiles.
<code>buffer</code>	character. users can either set <code>buffer</code> to <code>TRUE</code> , <code>FALSE</code> or provide buffer size in meters; if <code>TRUE</code> a predefined buffer is used where the size of the buffer depends on the lake size; if <code>FALSE</code> no buffer is applied; Default is <code>TRUE</code>
<code>edge</code>	character. Full path to GuidosToolbox Workbench MSPA tool see also (https://forest.jrc.ec.europa.eu/en/a)
<code>stream</code>	character. Full path to Hydrography90m stream network tif file add a reference were to check the necessary tif files for lake analysis

flow	character. Full path to Hydrography90m flow accumulation tif file
basins	character. Full path to Hydrography90m basin tif file
lake_dat	character. Full path of the output.csv table, i.e., the lake intersection table.
n_cores	interger. Number of cores used in parallelization, Default is one.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Author(s)

Jaime Garcia Marquez, Thomas Tomiczek

References

<https://grass.osgeo.org/grass82/manuals/> Soille P. and Vogt P. (2009). Morphological segmentation of binary patterns. Pattern Recognition Letters 30, 4:456-459, doi: 1.1016/j.patrec.2008.10.015
https://ies-ows.jrc.ec.europa.eu/gtb/GTB/MSPA_Guide.pdf

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Note: For this we need the lake IDs (i.e. HydroLake ID), extracted from the
# extract_lake_ids.R function; see also help(extract_lake_ids)

data <- fread(paste0(my_directory,
                     "/hydrography90m_test_data",
                     "/lake_id.txt"),
              header = TRUE)

lake_id <- "lake_id"

lakes <- (paste0(my_directory,
                 "/hydrography90m_test_data",
                 "/hydrography90m_test_lakes.gpkg"))

# To run the function we need to have installed GuidosToolbox Workbench MSPA tool
# see also (https://forest.jrc.ec.europa.eu/en/activities/lpa/gtb/) and
# (https://glowabio.github.io/hydrographr/articles/case\_study\_lake\_workflow.html)
# give full path virtual machine linux path
edge <- (paste0(/home/USER/GWB_version/,
                "/GWB"))

stream <- (paste0(my_directory,
                  "/hydrography90m_test_data",
                  "/stream_1264942.tif"))

flow <- (paste0(my_directory,
                "/hydrography90m_test_data",
                "/flow_1264942.tif"))
```

```

basins <- (paste0(my_directory,
                  "/hydrography90m_test_data",
                  "/basin_1264942.tif"))

lake_dat <- (paste0(my_directory,
                    "/hydrography90m_test_data"))

lake_intersect_table <- get_lake_intersection(data, lake_id = "lake_id",
                                             lakes, lake_name = "lake_id", buffer = TRUE, edge,
                                             stream, flow, basins, lake_dat, n_cores = 1,
                                             quiet = FALSE)

```

get_modelfit_table	<i>Get environmental variables at each occurrence and pseudo-absence point location</i>
--------------------	---

Description

Get the environmental variables for each species occurrences and pseudo-absences at given point locations by extracting the environmental information from the prediction table produced from the `get_predict_table` function see also `help(get_predict_table)`.

Usage

```

get_modelfit_table(
  data,
  spec,
  lon,
  lat,
  pseudoabs = 0,
  subc,
  predict_table,
  mod_fit_table,
  read = TRUE,
  quiet = TRUE
)

```

Arguments

data	a data.frame or data.table that contains the columns regarding the species name and the longitude / latitude coordinates in WGS84.
spec	character. The name of the column with the species names.
lon	character. The name of the column with the longitude coordinates.
lat	character. The name of the column with the latitude coordinates.
pseudoabs	integer. number of pseudo-absences
subc	character. Full path to the sub-catchment .tif file with the sub-catchment ID.

predict_table	character. Full path of the predict.csv table (i.e., output of get_predict_table); see also help(get_predict_table)
mod_fit_table	character. Full path of the output.csv table, i.e., the model fit table file.
read	logical. If TRUE, then the model .csv table gets read into R as data.table and data.frame. if FALSE, the table is only stored on disk. Default is FALSE.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Author(s)

Jaime Garcia Marquez, Thomas Tomiczek

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Load occurrence data

# Define full path to the sub-catchments raster layer

# Define full path to the prediction table
predict_tbl <- paste0(my_directory,
                      "/hydrography90m_test_data/projectionTB.csv")

# Define full path to the output model fit table
model_fit <- paste0(my_directory,
                    "/hydrography90m_test_data/model_table.csv")

# Get table with environmental variables at each occurrence
# and pseudo-absence point location
modelfit_table <- get_modelfit_table(data = species_occurrence,
                                     spec = "species",
                                     lon = "long",
                                     lat = "lat",
                                     pseudoabs = 10000,
                                     subc = subc_raster,
                                     predict_table = predict_tbl,
                                     mod_fit_table = model_fit)
```

get_pfafstetter_basins

Get Pfafstetter sub-basins

Description

Subset a basin or catchment into up to nine smaller sub-basins following the Pfafstetter basin delineation scheme. The function takes a network graph as the input and splits it into smaller sub-basins following a hierarchical topological coding scheme (see Verdin & Verdin (1999) for details), using the flow accumulation as the basis. The user has to define the sub-catchment (stream segment) ID that serves as the outlet of the basin. Note that this can be any stream segment that has an upstream catchment. The input graph can be created with [read_geopackage\(\)](#) and [get_catchment_graph\(\)](#).

Usage

```
get_pfafstetter_basins(
  g,
  subc_raster,
  out_dir,
  file_name,
  data_table = FALSE,
  n_cores = NULL
)
```

Arguments

<code>g</code>	igraph object. A directed graph of a basin with one outlet. The outlet can be any stream / sub-catchment for which the upstream basin should be split into smaller sub-basins. The input graph can be created with read_geopackage() and get_catchment_graph() .
<code>subc_raster</code>	character. Full path to the sub-catchment raster file of the basin. Does not need to be cropped / masked to the basin, but the IDs of the sub-catchments need to match with those in the input graph.
<code>out_dir</code>	character. The path of the output directory where the Pfafstetter raster layer will be written. Only needed when <code>data.table=FALSE</code> .
<code>file_name</code>	character. The filename and extension of the Pfafstetter raster layer (e.g. 'pfafstetter_raster.tif'). Only needed when <code>data.table=FALSE</code> .
<code>data_table</code>	Logical. If TRUE, then the result will be loaded into R as a 2-column data.table (sub-catchment ID and Pfafstetter code). If FALSE, the result is loaded as a raster (terra object) in R and written to disk. Default is FALSE.
<code>n_cores</code>	numeric. Number of cores used for parallelisation. Default is NULL (= <code>detectCores(logical=FALSE)-1</code>). Optional.

Value

Either a data.table, or a raster (terra object) loaded into R. In case the result is a raster, then a .tif file is written to disk.

Note

You can use the online map at <https://geo.igb-berlin.de/maps/351/view> to identify an ID of a stream segment (use the "Stream segment ID" layer to the left)

Author(s)

Sami Domisch

References

Verdin, K.L. & Verdin, J.P. (1999). A topological system for delineation and codification of the Earth's river basins. *Journal of Hydrology*, 218(1-2), 1-12. doi:10.1016/s0022-1694(99)00011-6

See Also

[read_geopackage\(\)](#) and [get_catchment_graph.\(\)](#) to create the input graph.

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Import the stream network as a graph
# Load stream network as a graph
my_graph <- read_geopackage(gpkg = paste0(my_directory,
                                           "/hydrography90m_test_data",
                                           "/order_vect_59.gpkg"),
                           import_as = "graph")

# Subset the graph such that it contains only one basin. You can use
# a random ID, i.e. it does not need to be the real outlet of the basin.
g_subset <- get_catchment_graph(g = my_graph,
                              subc_id = "513867227",
                              use_outlet = FALSE,
                              mode = "in",
                              as_graph = TRUE)

# Specify the sub-catchment raster file
subc_raster <- paste0(my_directory, "/hydrography90m_test_data",
                     "/subcatchment_1264942.tif")

# Specify the output directory
out_dir <- my_directory

# Calculate the Pfafstetter sub-basins and write the raster layer to disk (
# and import into R)
pfafstetter <- get_pfafstetter_basins(g = g_subset ,
                                     subc_raster = subc_raster,
                                     out_dir = out_dir,
                                     file_name = "pfafstetter_raster.tif",
                                     data_table = FALSE,
                                     n_cores = 4)
```

get_predict_table	<i>Get predict table</i>
-------------------	--------------------------

Description

This function creates a table by joining the information of one or more variables available in the Environment90m dataset. These variables need to be downloaded beforehand for the tiles overlapping with the area of interest. The subsetting of the original tables occurs by providing a list of subcatchment ids of interest.

Usage

```
get_predict_table(
  variable,
  statistics = "ALL",
  tile_id,
  input_var_path,
  subcatch_id,
  out_file_path,
  n_cores = NULL,
  read = TRUE,
  quiet = TRUE,
  tempdir = NULL,
  overwrite = FALSE
)
```

Arguments

statistics	character vector of statistics names. Possible values are "sd", "mean", "min", "max", "range" or "ALL". Default "ALL".
tile_id	character. The IDs of the tiles of interest.
input_var_path	path to directory that contains the tables of the Environment90m dataset. Tables may be in subdirectories of the provided directory.
subcatch_id	path to a text file with subcatchment ids, or numeric vector containing subcatchment ids.
out_file_path	character. The path to the output file to be created.
n_cores	numeric. Number of cores used for parallelization.
read	logical. If TRUE, the table with environmental variables gets read into R. If FALSE, the table is only stored on disk. Default is TRUE.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.
tempdir	String. Path to the directory where to store/look for the file size table. If not passed, defaults to the output of <code>base::tempdir()</code> .
overwrite	logical. If TRUE, the output file will be overwritten if it already exists. Useful for repeated testing. Default is FALSE.

variable(s) character vector of variable(s) names. Possible values are: all variables in the Environment90m dataset, which can be viewed by calling 'download__tables()'. For more details, see '?download_env90m_tables'.

Value

The function returns a table with

- subcatchment ids (subc_id)
- a column for each descriptive statistic of each variable (eg. bio1_mean: mean of the variable bio1)

Author(s)

Jaime García Márquez, Yusdiel Torres-Cambas

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Download one variable, in this case elevation, from the Environment90m dataset
# repository
download_merit_dem_tables(subset="ALL", tile_ids = "h18v04", download = TRUE,
download_dir = my_directory)

# prepare file with list subcatchments of interest
subc_raster <- paste0(my_directory, "subcatchment_1264942.tif")
target_ids <- extract_ids(subc_layer = subc_raster)
fwrite(target_ids, paste0(my_directory, 'subc_IDS.txt'))

# Define variable and tile:
var <- c("elev")
tile_id <- c("h18v04")

# Point to Env90m input data
in_path <- my_directory
# point to sub-catchment id table
subc_ids <- paste0(my_directory, '/hydrography90m_test_data/subc_IDS.txt')
# define output path
output <- paste0(my_directory, '/hydrography90m_test_data/predictTB.csv')

# Run the function with 2 cores
tb <- get_predict_table(variable = var,
                        statistics = c("ALL"),
                        tile_id = tile_id,
                        input_var_path = in_path,
                        subcatch_id = subc_ids,
                        out_file_path = output,
                        read = TRUE, quiet = FALSE,
```

```

n_cores = 2)

head(tb)

```

get_regional_unit_id *Get Hydrography90m regional unit IDs*

Description

Given the coordinates of input points (in WGS84), the function identifies the IDs of the regional units of the Hydrography90m in which the points are located. Input is a point data frame. The regional units refer to non-interrupted basins (as opposed to the 20°x20° tiles). These IDs can then be used to download the Hydrography90m regional unit raster mask(s) using [download_tiles\(\)](#).

Usage

```
get_regional_unit_id(data, lon, lat, quiet = TRUE, tempdir = NULL)
```

Arguments

data	a data.frame or data.table that contains the columns regarding the longitude / latitude coordinates in WGS84.
lon	character. The name of the column with the longitude coordinates.
lat	character. The name of the column with the latitude coordinates.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.
tempdir	character. The directory to be used as temp folder. Optional.

Author(s)

Afroditi Grigoropoulou

References

<https://gdal.org/programs/gdallocationinfo.html>

Examples

```

# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Read the species data
species_occurrence <- read.table(paste0(my_directory,
                                         "/hydrography90m_test_data",
                                         "/spdata_1264942.txt"),
                                header = TRUE)

```

```
# Get the regional unit ID
get_regional_unit_id(species_occurrence, lon = "longitude",
                     lat = "latitude")
```

```
get_segment_neighbours
```

Get stream segment neighbours

Description

For each input stream segment, the function reports those upstream, downstream, or up-and downstream segments that are connected to the input segment(s) within a specified neighbour order, with the option to summarize attributes across these segments. Note that the stream segment and sub-catchment IDs are identical, and for consistency, we use the term "subc_id". The input graph can be created with [read_geopackage\(\)](#) and [get_catchment_graph\(\)](#).

This function can be used to obtain the neighbour stream segments / sub-catchments for spatially explicit models or for spatial prioritization analyses (e.g. Marxan/Gurobi). By selecting a wider radius, the spatial dependency of the neighbouring segments / sub-catchments can be increased. See also [get_all_upstream_distances\(\)](#) which creates the input for addressing the longitudinal connectivity for the spatial prioritization.

Usage

```
get_segment_neighbours(
  g,
  subc_id = NULL,
  var_layer = NULL,
  stat = NULL,
  attach_only = FALSE,
  order = 5,
  mode = "in",
  n_cores = 1,
  max_size = 1500
)
```

Arguments

<code>g</code>	igraph object. A directed graph.
<code>subc_id</code>	numeric vector of the input sub-catchment IDs (=stream segment IDs) for which to search the connected segments.
<code>var_layer</code>	character vector. One or more attributes (variable layers) of the input graph that should be reported for each output segment_id ("to_stream"). Optional. Default is NULL.
<code>stat</code>	one of the functions mean, median, min, max, sd (without quotes). Aggregates (or summarizes) the variables for the neighbourhood of each input segment (e.g., the average land cover in the next five upstream segments or sub-catchments). Default is NULL.

attach_only	logical. If TRUE, the selected variables will be only attached to each segment without any further aggregation. Default is FALSE.
order	numeric. The neighbouring order as in <code>igraph::ego</code> . Order = 1 would be immediate neighbours of the input sub-catchment IDs, order = 2 would be the order 1 plus the immediate neighbours of those sub-catchment IDs in order 1, and so on.
mode	character. One of "in", "out", or "all". "in" returns only upstream neighbouring segments, "out" returns only the downstream segments, and "all" returns both.
n_cores	numeric. Number of cores used for parallelisation in the case of multiple stream segments / outlets. Default is 1. Note that the parallelisation process requires copying the data to each core. In case the graph is very large, and many segments are used as an input, setting <code>n_cores</code> to a higher value can speed up the computation. This comes however at the cost of possible RAM limitations and even slower processing since the large data will be copied to each core. Hence consider testing first with <code>n_cores = 1</code> . Optional.
max_size	numeric. Specifies the maximum size of the data passed to the parallel back-end in MB. Default is 1500 (1.5 GB). Consider a higher value for large study areas (more than one 20°x20° tile). Optional.

Value

A `data.table` indicating the connected segments (`stream | to_stream`), or a `data.table` that summarizes the attributes of those neighbours contributing to each segment.

#' @note Currently the attributes are not provided for the outlet (the selected `subc_id` segment). If the attributes are also needed for the outlet `subc_id`, then the next downstream `sub_id` can be selected (enlarge the study area) using e.g. `get_catchment_graph()`.

Author(s)

Sami Domisch

References

Csardi G, Nepusz T: The igraph software package for complex network research, InterJournal, Complex Systems 1695. 2006. <https://igraph.org>

See Also

`read_geopackage()` and `get_catchment_graph.()` to create the input graph. The function `get_all_upstream_distances` can be used to obtain the network distance for all upstream `subc_id` (useful as an input for a subsequent spatial prioritization analysis).

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)
```

```

# Load the stream network as graph
my_graph <- read_geopackage(gpkg= paste0(my_directory,
                                         "/hydrography90m_test_data",
                                         "/order_vect_59.gpkg"),
                           import_as = "graph")

# Subset the graph and get a smaller catchment
my_graph <- get_catchment_graph(g = my_graph, subc_id = 513866048, mode = "in",
                               use_outlet = FALSE, as_graph = TRUE, n_cores = 1)

# Get a vector of all segment IDs
library(igraph)
subc_id <- as_ids(V(my_graph))

# Get all (up-and downstream) directly adjacent neighbours of each segment
get_segment_neighbours(g = my_graph, subc_id = subc_id,
                       order = 1, mode = "all")

# Get the upstream segment neighbours in the 5th order
# and report the length and source elevation
# for the neighbours of each input segment
get_segment_neighbours(g = my_graph, subc_id = subc_id,
                       order = 5, mode = "in", n_cores = 1,
                       var_layer = c("length", "source_elev"),
                       attach_only = TRUE)

# Get the downstream segment neighbours in the 5th order
# and calculate the median length and source elevation
# across the neighbours of each input segment
get_segment_neighbours(g = my_graph, subc_id = subc_id,
                       order = 2, mode = "out", n_cores = 1,
                       var_layer = c("length", "source_elev"),
                       stat = median)

# Get the up-and downstream segment neighbours in the 5th order
# and report the median length and source elevation
# for the neighbours of each input segment
get_segment_neighbours(g = my_graph, subc_id = subc_id, order = 2,
                       mode = "all", n_cores = 1,
                       var_layer = c("length", "source_elev"),
                       stat = mean, attach_only = TRUE)

```


Description

Identifies the 20°x20° tile IDs of the Hydrography90m data in which the input points are located. The IDs can then be used to download the data using `download_tiles()`. The input is a data frame with point coordinates. For orientation, please also see the tiles at the https://hydrography.org/hydrography90m/hydrography90m_layers

Usage

```
get_tile_id(data, lon, lat, tempdir = NULL)
```

Arguments

<code>data</code>	a data.frame or data.table that contains the columns regarding the longitude / latitude coordinates in WGS84.
<code>lon</code>	character. The name of the column with the longitude coordinates.
<code>lat</code>	character. The name of the column with the latitude coordinates.
<code>tempdir</code>	character. The directory to be used as temp folder. Optional.

Author(s)

Afroditi Grigoropoulou

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Load species occurrence data
species_occurrence <- read.table(paste0(my_directory,
                                         "/hydrography90m_test_data",
                                         "/spdata_1264942.txt"),
                                header = TRUE)

# Get the tile ID
get_tile_id(data = species_occurrence,
            lon = "longitude", lat = "latitude")
```

```
get_upstream_catchment
```

Delineate the upstream catchment

Description

Delineates the upstream catchment of a given point, where the point is considered as the outlet of the catchment.

Usage

```

get_upstream_catchment(
  data,
  id,
  lon,
  lat,
  direction_layer = NULL,
  out_dir = NULL,
  n_cores = NULL,
  compression = "low",
  bigtiff = TRUE,
  quiet = TRUE
)

```

Arguments

<code>data</code>	a data.frame or data.table that contains the columns regarding the longitude / latitude coordinates in WGS84. Note that the points need to be snapped to the stream network with snap_to_network() or snap_to_subc_segment() .
<code>id</code>	character. The name of a column containing unique IDs for each row of "data" (e.g., occurrence or site IDs). The unique IDs need to be numeric and less than 10 characters long.
<code>lon</code>	character. The name of the column with the longitude coordinates.
<code>lat</code>	character. The name of the column with the latitude coordinates.
<code>direction_layer</code>	character. Full path to the flow direction raster file.
<code>out_dir</code>	Full path to the directory where the output(s) will be stored. The output file name includes the "id" which helps identifying the upstream corresponding catchment.
<code>n_cores</code>	numeric. Number of cores used for parallelisation. If NULL, available cores - 1 will be used. Default is NULL.
<code>compression</code>	character. Compression of the written output file. Compression levels can be defined as "none", "low", or "high". Default is "low", referring to compression type "DEFLATE" and compression level 2. "high" refers to compression level 9.
<code>bigtiff</code>	logical. Define whether the output file is expected to be a BIGTIFF (file size larger than 4 GB). If FALSE and size > 4GB no file will be written. Default is TRUE.
<code>quiet</code>	logical. If FALSE, the standard output will be printed. Default is TRUE.

Author(s)

Jaime Garcia Marquez, Afroditi Grigoropoulou, Marlene Schürz

References

- <https://grass.osgeo.org/grass82/manuals/r.water.outlet.html>
- <https://grass.osgeo.org/grass82/manuals/r.region.html>

See Also

- [snap_to_network](#) to snap the data points to the next stream segment within a given radius and/or a given flow accumulation threshold value.
- [snap_to_subc_segment](#) to snap the data points to the next stream segment within the sub-catchment the point is located.
- [extract_ids](#) to extract basin and sub-catchment IDs.

Examples

```
# Download test data into temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Before running the function get_upstream_catchment(), snap the points to
# to the stream segment. There are multiple ways to snap the points. Here is
# one example:

# Load occurrence data
species_occurrence <- read.table(paste0(my_directory,
                                         "/hydrography90m_test_data",
                                         "/spdata_1264942.txt"),
                                header = TRUE)

# Define full path to the basin and sub-catchments raster layer
basin_raster <- paste0(my_directory,
                      "/hydrography90m_test_data/basin_1264942.tif")
subc_raster <- paste0(my_directory,
                     "/hydrography90m_test_data/subcatchment_1264942.tif")

# Define full path to the vector file of the stream network
stream_vector <- paste0(my_directory,
                       "/hydrography90m_test_data/order_vect_59.gpkg")

# Automatically extract the basin and sub-catchment IDs and
# snap the data points to the stream segment
snapped_coordinates <- snap_to_subc_segment(data = species_occurrence,
                                           lon = "longitude",
                                           lat = "latitude",
                                           id = "occurrence_id",
                                           basin_layer = basin_raster,
                                           subc_layer = subc_raster,
                                           stream_layer = stream_vector,
                                           n_cores = 2)

# Define full path to the direction .tif
direction_raster <- paste0(my_directory,
                          "/hydrography90m_test_data/direction_1264942.tif")

# Define the path for the output file(s)
output_folder <- paste0(my_directory, "/upstream_catchments")
if(!dir.exists(output_folder)) dir.create(output_folder)
```

```
# Get the upstream catchment for each point location
get_upstream_catchment(snapped_coordinates,
                        lon = "lon_snap",
                        lat = "lat_snap",
                        id = "occurrence_id",
                        direction_layer = direction_raster,
                        out_dir = output_folder,
                        n_cores = 2)
```

get_upstream_variable *Calculate upstream variables for each sub-catchment*

Description

For each input sub-catchment, the function calculates the upstream mean, median, min, max, sd or sum for one or more variables. Note that the stream segment and sub-catchment IDs are identical. The input graph can be created with [read_geopackage\(\)](#) and [get_catchment_graph\(\)](#).

This function can be used to obtain all upstream stream segments / sub-catchments for species distribution modelling, or for spatial prioritization analyses (e.g. Marxan/Gurobi) to specify the connectivity (and in this case the "length" attribute can be used).

The function accepts as an input a graph with one or more outlets (e.g. an entire tile with many drainage basins). The variable that should be aggregated upstream should be prepared as a data.table, e.g. with [extract_zonal_stat\(\)](#).

Usage

```
get_upstream_variable(
  g,
  variable_table = NULL,
  subc_id = NULL,
  var_layer = NULL,
  upstream_stat = NULL,
  include_focal = FALSE,
  save_up_conn = NULL,
  load_up_conn = NULL,
  n_cores = 1,
  max_size = 3000
)
```

Arguments

g	igraph object. A directed graph.
variable_table	a data.table that includes the stream column (corresponding to the subc_id) as well as the attributes that should be aggregated across the the upstream network subc_id. Default is NULL.
subc_id	A vector of subc_id for which the upstream variables should be calculated. Optional; default is to use all subc_id of the input graph.

var_layer	character vector. One or more attributes (variable layers) of the variable_table should be reported for each output subc_id. Default is NULL.
upstream_stat	one of the functions mean, median, min, max, sd, sum (without quotes). The function will be used to aggregate (or summarize) the upstream variables for each subc_id (e.g., the average land cover across the entire upstream area). Default is NULL.
include_focal	Whether the focal subc_id should be included in the aggregation with include_focal = TRUE which is the default. Set to FALSE if the focal subc_id should not be included in the upstream aggregation of the given sub-catchment.
save_up_conn	character. Provide a name of the .RData file that will be written to disk (to getwd()), and which includes the intermediate result consisting of a data.table that includes all upstream connections for each subc_id. Useful for large study areas as it avoids re-running the possibly time-consuming pre-processing to obtain other metrics (e.g. mean, sum) or other variables. The data.table is called upstream_dt and has the columns stream and base. Default is FALSE.
load_up_conn	Optional, and if used, it should be upstream_dt. In case the file with the upstream connections was previously written to disk (using e.g. save_up_conn = "my_file"), then this data.table can be first loaded with load(paste0(getwd(), "/my_file.RData")), which loads the upstream_dt data.table with the columns c("stream", "base"). Use load_up_conn = upstream_dt to then skip the pre-processing. Optional, default is NULL.
n_cores	numeric. Number of cores used for parallelisation. In case the graph is very large, and many segments are used as an input, setting n_cores to a higher value can speed up the computation. Note however that the parallelisation process requires copying the input graph to each core. This may result in possible RAM limitations and even slower processing. Hence consider testing first with a lower number of cores. Default is n_cores = 1. Optional.
max_size	numeric. Specifies the maximum size of the data passed to the parallel back-end in MB. Default is 1500 (1.5 GB). Consider a higher value for large study areas (more than one 20°x20° tile). Optional.

Value

A data.table indicating the "stream" (=subc_id) and the upstream variables which have the same column names as the var_layer argument. The intermediate output can be saved to disk (requires setting save_up_conn = "my_file").

Author(s)

Sami Domisch

References

Csardi G, Nepusz T: The igraph software package for complex network research, InterJournal, Complex Systems 1695. 2006. <https://igraph.org>

See Also

[read_geopackage\(\)](#) and [get_catchment_graph\(\)](#) to create the input graph. Alternatively, see [get_segment_neighbours\(\)](#) to obtain the upstream variables for a specified neighbourhood.

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
setwd(my_directory)
download_test_data(my_directory)

# Load the stream network as graph
my_graph <- read_geopackage(gpkg = paste0(my_directory,
                                           "/hydrography90m_test_data",
                                           "/order_vect_59.gpkg"),
                           import_as = "graph")

# Subset the graph and get a smaller catchment
my_graph <- get_catchment_graph(g = my_graph,
                               subc_id = 513867228,
                               mode = "in",
                               use_outlet = FALSE,
                               as_graph = TRUE,
                               n_cores = 1)

## Prepare the variables that should be accumulated.
## Load the table
variable_table <- read_geopackage(gpkg = paste0(my_directory,
                                                  "/hydrography90m_test_data",
                                                  "/order_vect_59.gpkg"),
                                  import_as = "data.table")

## Specify the layers for the upstream aggregation
var_layer = c("length", "flow_accum")

## Subset the table
keep_these <- c("stream", var_layer)
variable_table <- variable_table[, ..keep_these]

## Get the upstream sum of the variables "length" and "flow_accum" for
## single subc_id
result <- get_upstream_variable(my_graph,
                                variable_table = variable_table,
                                var_layer = var_layer,
                                upstream_stat = sum,
                                subc_id = c(513861908, 513864129),
                                include_focal = TRUE)
```

```

## Get the upstream sum of the variables "length" and "flow_accum" across the
## entire network
result <- get_upstream_variable(my_graph,
                              variable_table = variable_table,
                              var_layer = var_layer,
                              upstream_stat=sum,
                              include_focal = TRUE,
                              n_cores = 4,
                              save_up_conn = "my_file")

## Alternatively, load the previously generated upstream connections
## and use it directly, skipping the pre-processing
load(paste0(getwd(), "/my_file.RData"))

result <- get_upstream_variable(my_graph,
                              variable_table = variable_table,
                              var_layer = var_layer,
                              upstream_stat=max,
                              include_focal = TRUE,
                              load_up_conn = upstream_dt)

## Map the new variable across the network

## Specify tif-layer for reclassification
subc_raster <- paste0(my_directory, "/hydrography90m_test_data/subcatchment_1264942.tif")
recl_raster <- paste0(my_directory, "/upstream_sum.tif")

## Set columns as integer
result <- result[, names(result) := lapply(.SD, as.integer)]

### Create raster - select the "to" column which represents the unique subc_id
r <- reclass_raster(data = result,
                   rast_val = "stream",
                   new_val = "flow_accum",
                   raster_layer = subc_raster,
                   recl_layer = recl_raster,
                   read = TRUE)

## Plot the map
terra::plot(r, background = "grey")

```

Description

Merge multiple raster or spatial vector objects from disk to form a new raster or spatial vector object with a larger spatial extent. A directory with at least two raster .tif or spatial vector geopackage files should be provided. Depending on the input, the output is a .tif or a .gpkg file (saved under out_dir). If read = TRUE, the output is read into R as a SpatRaster (terra package) object in case of .tif files, or as a SpatVector (terra package) object in case of .gpkg files.

Usage

```
merge_tiles(
  tile_dir,
  tile_names,
  out_dir,
  file_name,
  name = "stream",
  compression = "low",
  bigtiff = TRUE,
  read = FALSE,
  quiet = TRUE
)
```

Arguments

tile_dir	character. The directory containing the raster or spatial vectors tiles, which should be merged.
tile_names	character. The names of the files to be merged, including the file extension (.tif or .gpkg).
out_dir	character. The directory where the output will be stored.
file_name	character. Name of the merged output file, including the file extension (.tif or .gpkg).
name	character. The attribute table column name of the stream segment ("stream"), sub-catchment ("ID"), basin ("ID") or outlet ("ID") column which is used for merging GeoPackages. Default is "stream".
compression	character. Compression of the written output file. Compression levels can be defined as "none", "low", or "high". Default is "low", referring to compression type "DEFLATE" and compression level 2. "high" refers to compression level 9.
bigtiff	logical. Define whether the output file is expected to be a BIGTIFF (file size larger than 4 GB). If FALSE and size > 4GB no file will be written. Default is TRUE.
read	logical. If TRUE, the merged layer gets read into R. If FALSE, the layer is only stored on disk. Default is FALSE.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Value

A .tif raster file or .gpkg spatial vector object that is always written to disk, and optionally loaded into R.

Author(s)

Thomas Tomiczek, Jaime Garcia Marquez, Afroditi Grigoropoulou

References

<https://gdal.org/programs/gdalbuildvrt.html>
https://gdal.org/programs/gdal_translate.html
<https://gdal.org/programs/ogrmerge.html#ogrmerge>
<https://gdal.org/programs/ogr2ogr.html>

Examples

```
# Download tiles into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_tiles(variable = "basin",
               file_format = "tif",
               tile_id = c("h22v08", "h22v10"),
               download_dir = my_directory)

# Define folder containing only the tiles, which should be merged
tiles_folder <- paste0(my_directory, "/r.watershed/basin_tiles20d")
# Define output folder
output_folder <- paste0(my_directory, "/merged_tiles")
# Create output folder if it doesn't exist
if(!dir.exists(output_folder)) dir.create(output_folder)

# Merge tiles
merged_tiles <- merge_tiles(tile_dir = tiles_folder,
                           tile_names = c("basin_h22v08.tif", "basin_h22v10.tif"),
                           out_dir = output_folder,
                           file_name = "basin_merged.tif",
                           read = TRUE)
```

read_geopackage

Read a GeoPackage file

Description

Reads an entire, or a subset of a GeoPackage vector file from disk either as a table (data.table), as a directed graph object (igraph), a spatial dataframe (sf) or a SpatVect object (terra).

Usage

```
read_geopackage(
  gpkg,
  import_as = "data.table",
  layer_name = NULL,
  subc_id = NULL,
  name = "stream"
)
```

Arguments

<code>gpkg</code>	character. Full path of the GeoPackage file.
<code>import_as</code>	character. "data.table", "graph", "sf", or "SpatVect". "data.table" imports data as a data.table. "graph" imports the layer as a directed graph (igraph object). This option is only possible for a network layer (e.g. the stream network) and it needs to contain the attributes "stream" and "next_stream". "sf" imports the layer as a spatial data frame (sf object). "SpatVect" imports the layer as a SpatVector (terra object). Default is "data.table".
<code>layer_name</code>	character. Name of the specific data layer to import from the GeoPackage. A specific data layer only needs to be defined if the GeoPackage contains multiple layers. To see the available layers the function <code>st_layers()</code> from the R package 'sf' can be used. Optional. Default is NULL.
<code>subc_id</code>	numeric. Vector of the sub-catchment (or stream segment) IDs in the form of <code>(c(ID1, ID2, ...))</code> for which the spatial objects or attributes of the GeoPackage should be imported. Optional. Default is NULL.
<code>name</code>	character. The attribute table column name of the stream segment ("stream"), sub-catchment ("ID"), basin ("ID") or outlet ("ID") column which is used for subsetting the GeoPackage prior importing. Optional. Default is "stream".

Author(s)

Sami Domisch, Marlene Schürz

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Read the stream network as a graph
my_graph <- read_geopackage(gpkg = paste0(my_directory,
                                           "/hydrography90m_test_data",
                                           "/order_vect_59.gpkg"),
                           import_as = "graph")

# Read the stream network as a data.table
my_dt <- read_geopackage(gpkg = paste0(my_directory,
```

```

"/hydrography90m_test_data",
"/order_vect_59.gpkg"))

# Read the stream network as a data.table for specific IDs
my_dt <- read_geopackage(gpkg = paste0(my_directory,
"/hydrography90m_test_data",
"/order_vect_59.gpkg"),
subc_id = c(513833203, 513833594))

# Read the sub-catchments as a SF-object
my_sf <- read_geopackage(gpkg = paste0(my_directory,
"/hydrography90m_test_data",
"/sub_catchment_59.gpkg"),
import_as = "sf",
layer_name = "sub_catchment")

# Read a subset of sub-catchments as a SF-object
my_sf <- read_geopackage(gpkg = paste0(my_directory,
"/hydrography90m_test_data",
"/sub_catchment_59.gpkg"),
import_as = "sf",
subc_id = c(513833203, 513833594),
name = "ID")

# Read the basin as SpatVect object
my_sv <- read_geopackage(gpkg = paste0(my_directory,
"/hydrography90m_test_data",
"/basin_59.gpkg"),
import_as = "SpatVect")

```

reclass_raster	<i>Reclassify a raster layer</i>
----------------	----------------------------------

Description

Reclassifies a raster .tif layer based on a look-up table, such that the output raster contains the new values. The function uses the `r.reclass` function of GRASS GIS.

Note that the input raster needs to be of type integer. If the input raster layer has floating point values, you can multiply it by some factor (e.g. 1000) to achieve integer values, otherwise the GRASS GIS `r.reclass` will round the raster values down to the next integer which is not always desired.

Usage

```

reclass_raster(
  data = NULL,
  rast_val = NULL,
  new_val = NULL,

```

```

    remaining = NULL,
    remaining_value = -9999,
    raster_layer,
    recl_layer,
    no_data = -9999,
    type = "Int32",
    compression = "low",
    bigtiff = TRUE,
    read = FALSE,
    quiet = TRUE
  )

```

Arguments

data	a data.frame or data.table with the original and new values to be written to the raster.
rast_val	character. The name of the column with the original raster values.
new_val	character. The name of the column with the new raster values, which need to be integer values. In case of floating point values, consider multiplying the values e.g. by 1000 to keep three decimals.
remaining	character. How to treat raster values not listed in the reclassification table: "same" retains their original values (equivalent to '* = *' in GRASS), "value" assigns a fixed value ('remaining_value'), and 'NULL' (default) does nothing. When 'remaining = "same"', 'remaining_value' is overlooked.
remaining_value	numeric. Value to assign if 'remaining = "value"'. Default is -9999.
raster_layer	Full path to the input raster .tif layer.
recl_layer	character. Full path of the output .tif layer, i.e., the reclassified raster file.
no_data	numeric. The no_data value of the new .tif layer. Default is -9999.
type	character. Data type; Options are Byte, Int16, UInt16, Int32, UInt32, CInt16, CInt32. Default is Int32. Note that only integer values are allowed.
compression	character. Compression of the written output file. Compression levels can be defined as "none", "low", or "high". Default is "low", referring to compression type "DEFLATE" and compression level 2. "high" refers to compression level 9.
bigtiff	logical. Define whether the output file is expected to be a BIGTIFF (file size larger than 4 GB). If FALSE and size > 4GB no file will be written. Default is TRUE.
read	logical. If TRUE, then the reclassified raster .tif layer gets read into R as a SpatRaster (terra object). If FALSE, the layer is only stored on disk. Default is FALSE.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Author(s)

Marlene Schürz, Afroditi Grigoropoulou


```
recl_layer = recl_raster)
```

report_no_data	<i>Report NoData value</i>
----------------	----------------------------

Description

This function reports the defined NoData value of a raster layer. The NoData value of a raster layer represents the absence of data. In computations the NoData value can be treated in different ways. Either the NoData value will be reported or the Nodata value will be ignored and a value is computed from the available values of a specified location.

Usage

```
report_no_data(data_dir, var_layer, n_cores = NULL)
```

Arguments

data_dir	character. Path to the directory containing all input data.
var_layer	character vector of variable raster layers on disk, e.g. "slope_grad_dw_cel_h00v00.tif".
n_cores	numeric. Number of cores used for parallelization, in case multiple .tif files are provided to var_layer.

Author(s)

Afroditi Grigoropoulou, Marlene Schürz

References

<https://gdal.org/programs/gdalinfo.html>

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Report the NoData value
report_no_data(data_dir = paste0(my_directory, "/hydrography90m_test_data"),
               var_layer = c("subcatchment_1264942.tif", "flow_1264942.tif",
                             "spi_1264942.tif"),
               n_core = 2)
```

set_no_data	<i>Set no data value</i>
-------------	--------------------------

Description

Change or set the NoData value for a raster layer. The change happens in-place, meaning that the original file is overwritten on disk.

Usage

```
set_no_data(data_dir, var_layer, no_data, quiet = TRUE)
```

Arguments

data_dir	character. Path to the directory containing all input data.
var_layer	character vector of variable layers on disk, e.g. c("sti_h16v02.tif", "slope_grad_dw_cel_h00v00.tif"). The original files will be overwritten.
no_data	numeric. The desired NoData value.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Author(s)

Afroditi Grigoropoulou, Marlene Schürz

References

https://gdal.org/programs/gdal_edit.html

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Define no data value
set_no_data(data_dir = paste0(my_directory, "/hydrography90m_test_data"),
            var_layer = "cti_1264942.tif",
            no_data = -9999)
```

snap_to_network	<i>Snap points to stream segment based on distance or flow accumulation</i>
-----------------	---

Description

Snap points to the next stream segment within a defined radius (in map pixels) or a minimum flow accumulation.

Usage

```
snap_to_network(
  data,
  lon,
  lat,
  id,
  stream_layer,
  accu_layer = NULL,
  method = "distance",
  distance = 500,
  accumulation = 0.5,
  quiet = TRUE
)
```

Arguments

data	a data.frame or data.table that contains the columns regarding the longitude / latitude coordinates in WGS84.
lon	character. The name of the column with the longitude coordinates.
lat	character. The name of the column with the latitude coordinates.
id	character. The name of a column containing unique IDs for each row of "data" (e.g., occurrence or site IDs). The unique IDs need to be numeric and less than 10 characters long.
stream_layer	character. Full path of the stream network .tif file
accu_layer	character. Full path of the flow accumulation .tif file. Needed if the point should be snapped to the next stream segment having an accumulation value higher than the flow accumulation threshold (set by 'accumulation'). This prevents points from being snapped to small stream tributaries. Optional. Default is NULL.
method	character. One of "distance", "accumulation", or "both". Defines if the points are snapped using the distance or flow accumulation (see "Details" for more information). If method is set to "both" the output will contain the new coordinates for both calculations. Default is "distance" (in map pixels).
distance	numeric. Maximum radius in map pixels. The points will be snapped to the next stream within this radius. Default is 500.

accumulation	numeric. Minimum flow accumulation. Points will be snapped to the next stream with a flow accumulation equal or higher than the given value. Default is 0.5.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Details

The function makes use of the `r.stream.snap` function available in GRASS GIS to simultaneously snap a number of points to the stream network. A distance threshold can be specified and points will be snapped to any stream segment within this distance radius (in map pixels). However, to avoid snapping to small tributaries, an accumulation threshold can be used and the snapping occurs on stream segment with equal or higher accumulation threshold and within the given distance radius.

Value

Returns a data.frame with the snapped coordinates and the sub-catchment ID of the snapped stream segment. If the sub-catchment ID is NA, no stream segment was found within the given distance (method = "distance") or no stream segment was found within the given distance and a flow accumulation equal or higher than the given threshold (method = "accumulation"). "out-bbox" means that the provided coordinates are not within the extend (bounding box) of the provided stream network layer.

Note

Duplicated rows will be removed from the input data.

Author(s)

Marlene Schürz, Jaime Garcia Marquez

References

<https://grass.osgeo.org/grass78/manuals/addons/r.stream.snap.html>

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Load occurrence data
species_occurrence <- read.table(paste0(my_directory,
                                          "/hydrography90m_test_data/spdata_1264942.txt"),
                                header = TRUE)

# Define full path to stream network and flow accumulation
stream_raster <- paste0(my_directory,
                        "/hydrography90m_test_data/stream_1264942.tif")
flow_raster <- paste0(my_directory,
                     "/hydrography90m_test_data/flow_1264942.tif")
```

```
# To calculate the new (snapped) coordinates for a radius and a flow
snapped_coordinates <- snap_to_network(data = species_occurrence,
                                       lon = "longitude",
                                       lat = "latitude",
                                       id = "occurrence_id",
                                       stream_layer = stream_raster,
                                       accu_layer = flow_raster,
                                       method = "both",
                                       distance = 300,
                                       accumulation = 0.8)

# Show head of output table
head(snapped_coordinates)
```

snap_to_subc_segment *Snap points to stream segment within the sub-catchment*

Description

Move points to the stream segment within the sub-catchment where the point is located.

Usage

```
snap_to_subc_segment(
  data,
  lon,
  lat,
  id,
  basin_id = NULL,
  subc_id = NULL,
  basin_layer,
  subc_layer,
  stream_layer,
  n_cores = 1,
  quiet = TRUE
)
```

Arguments

data	a data.frame or data.table that contains the columns regarding the longitude / latitude coordinates in WGS84.
lon	character. The name of the column with the longitude coordinates.
lat	character. The name of the column with the latitude coordinates.
id	character. The name of a column containing unique IDs for each row of "data" (e.g., occurrence or site IDs). The unique IDs need to be numeric and less than 10 characters long.

basin_id	character. The name of the column with the basin IDs. If NULL, the basin IDs will be extracted automatically. Optional. Default is NULL
subc_id	character. The name of the column with the sub-catchment IDs. If NULL, the sub-catchment IDs will be extracted automatically. Optional. Default is NULL.
basin_layer	character. Full path to the basin ID .tif layer.
subc_layer	character. Full path to the sub-catchment ID .tif layer.
stream_layer	character. Full path of the stream network .gpkg file.
n_cores	numeric. Number of cores used for parallelisation. Default is 1.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Details

The function uses the network module of GRASS GIS (v.net), to connect a vector line map (stream network) with a point map (occurrence/sampling points). After masking the stream segment and the sub-catchment where the target point is located, the connect operation snaps the point to the stream segment using a distance threshold. This threshold is automatically calculated as the longest distance between two points within the sub-catchment. In this way the snapping will always take place. From the new location, the function extracts the new snapped coordinates.

Value

A data.table of the original and new coordinates, along with the sub-catchment ID.

Author(s)

Jaime Garcia Marquez, Marlene Schürz

References

<https://grass.osgeo.org/grass82/manuals/v.net.html>

See Also

- `snap_to_network()` to snap the data points to the next stream segment within a given radius and/or a given flow accumulation threshold value.
- `extract_ids()` to extract basin and sub-catchment IDs.

Examples

```
# Download test data into the temporary R folder
# or define a different directory
my_directory <- tempdir()
download_test_data(my_directory)

# Load occurrence data
species_occurrence <- read.table(paste0(my_directory,
                                         "/hydrography90m_test_data/spdata_1264942.txt"),
                                header = TRUE)
```

```

basin_rast <- paste0(my_directory,
                    "/hydrography90m_test_data/basin_1264942.tif")
subc_rast <- paste0(my_directory,
                   "/hydrography90m_test_data/subcatchment_1264942.tif")

# Define full path to the vector file of the stream network
stream_vect <- paste0(my_directory,
                     "/hydrography90m_test_data/order_vect_59.gpkg")

hydrography90m_ids <- extract_ids(data = species_occurrence,
                                lon = "longitude",
                                lat = "latitude",
                                id = "occurrence_id",
                                subc_layer = subc_rast,
                                basin_layer = basin_rast)

# Snap data points to the stream segment of the provided sub-catchment ID
snapped_coordinates <- snap_to_subc_segment(data = hydrography90m_ids,
                                           lon = "longitude",
                                           lat = "latitude",
                                           id = "occurrence_id",
                                           basin_id = "basin_id",
                                           subc_id = "subcatchment_id",
                                           basin_layer = basin_rast,
                                           subc_layer = subc_rast,
                                           stream_layer = stream_vect,
                                           n_cores = 2)

# Show head of output table
head(snapped_coordinates)

# OR
# Automatically extract the basin and sub-catchment IDs and
# snap the data points to the stream segment
snapped_coordinates <- snap_to_subc_segment(data = species_occurrence,
                                           lon = "longitude",
                                           lat = "latitude",
                                           id = "occurrence_id",
                                           basin_layer = basin_rast,
                                           subc_layer = subc_rast,
                                           stream_layer = stream_vect,
                                           n_cores = 2)

# Show head of output table
head(snapped_coordinates)

```

split_table

Split table

Description

Split the table along a set number of rows into multiple parts of equal length.

Note that duplicated rows will be removed.

Usage

```
split_table(data, split = NULL, split_tbl_path, read = FALSE, quiet = TRUE)
```

Arguments

data	a data.frame or data.table
split	numeric. number of rows selected to split the table
split_tbl_path	character. Full path to store the split tables
read	logical. If TRUE, then the split data tables get read into R as a data tables. If FALSE, the tables are only stored on disk. Default is FALSE.
quiet	logical. If FALSE, the standard output will be printed. Default is TRUE.

Author(s)

Jaime Garcia Marquez, Thomas Tomiczek

Examples

```
# Create data table

df <- data.frame(matrix(ncol = 2, nrow = 10000))
colnames(df) <- c('var1', 'var2')
# or with real data
my_directory <- tempdir()
download_test_data(my_directory)
df <- fread(paste0(my_directory, '/projectionTB.csv'), fill=TRUE)

# Define full path to store split data tables
split_tbl_path <- paste0(my_directory,
                          "/hydrography90m_test_data/")

# Split data table
hydrography90m_ids <- split_table(df, split = 20000, split_tbl_path)

# Show the output table
hydrography90m_ids
```

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